Chapter 5

Module III Market

Market is a place or a process where the interaction between buyers and sellers takes place in order to buy or sell a product. This interaction may take place at a particular place, over telephone or through internet. There are different types of market structures in an economy. It depends on nature of competition, type of product, number of buyers and sellers, freedom of entry and exit from the market etc. Based on these features a market can be perfect competition, monopoly, monopolistic competition, oligopoly etc.

5.1 Perfect Competition

Perfect competition is a market situation in which there are a large number of buyers and sellers dealing in a homogeneous product with perfect knowledge about the market conditions and perfect mobility of goods and factors of production. It appears to be a hypothetical market condition but the study of this market situation is very essential to understand other forms of market structures. The following are the important features of perfect competition.

1. **Large number of buyers and sellers** – The number of buyers and sellers is so large that the act of a single seller or buyer cannot influence price or output in the market. That is each seller or buyer is an insignificant part of the market.
2. **Homogeneous product** – Under perfect competition all sellers are selling an identical product which is same in appearance, colour, quality etc. and hence they are perfect substitutes. Since all sellers are selling the same product, they can charge only the same price.
3. **Freedom of entry and exit** – There are no restrictions on the entry and exit of firms. Thus there is open competition. It ensures normal profit in the market. If the existing firms are earning supernormal profit new firms will enter into the market. This increases supply and reduces the profit margin. On the other hand if there is loss some of the sellers will leave the market. This help the remaining sellers to earn normal profit.
4. **Perfect knowledge** – Buyers and sellers have perfect knowledge about market conditions. Buyers know about the product and its price. Similarly

sellers have the knowledge about the price of the factor inputs, technology etc. This also ensures a uniform price of the product.

5. **Perfect mobility of goods and factors of production** – Goods and factors of production are free to move from one place to another place or from one industry to another industry.
6. **Absence of transport cost** – It is assumed that transport cost is absent in perfect competition. This also ensures uniform price.
7. **Perfectly elastic demand curve** – Under perfect competition a firm can sell any amount of the commodity at the price determined in the market by market demand and market supply. Hence the demand curve is a horizontal straight line parallel to the x-axis.

Equilibrium of a Firm

Under any market situation a firm is in equilibrium when it gets maximum profit.

There are two approaches to find the profit maximising level of output.

TC, TR Approach

Under this approach a firm will be in equilibrium when it produce that level of output where the difference between TR and TC(profit) is the maximum.

In the diagram when the firm produces Q level of output the difference between TR and TC is the maximum. Hence, it is the equilibrium output. To find the equilibrium output a tangent is drawn to the TC curve which is parallel to the TR curve. This is the case in perfect competition.

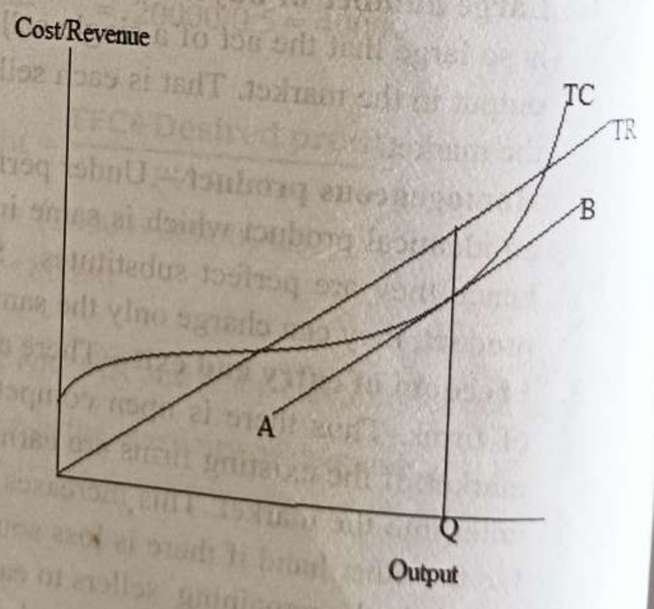


Fig 5.1 TC,TR Approach

MC MR Approach

Under this approach profit will be maximum when the following two conditions are satisfied.

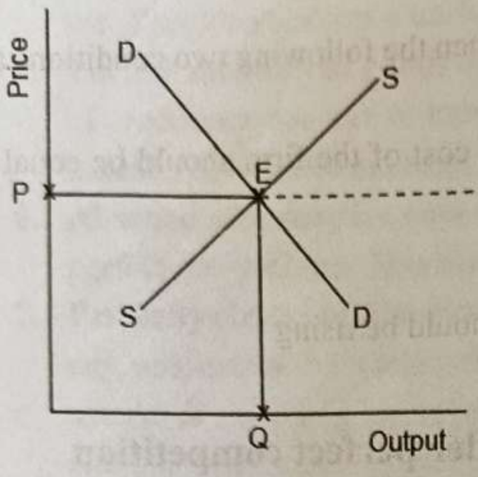
1. At the equilibrium point marginal cost of the firm should be equal to its marginal revenue (MC = MR)
2. At the point of equilibrium MC should be rising

AR curve and MR curve of a firm under perfect competition

Under perfect competition price of a product is determined for the entire industry by the forces of market demand and market supply. This price is accepted by each firm in the industry. Therefore a seller under perfect competition is called a price taker. A seller can sell any amount of the commodity at this price. Hence the demand curve facing a seller under perfect competition is perfectly elastic. It is a horizontal straight line parallel to the x-axis. If he increases the price he will lose his customers because all the sellers are selling an identical product. If he reduces the price he will face a loss because under perfect competition usually a firm get normal profit.

> In Economics a firm means a single business establishment which produces a product. Industry comprises all such firms produce the same product in the market.

A seller can sell any amount at the price prevailing in the market. Whether he sells one unit or thousand unit, price will be the same. Since all units of the commodity are sold at the same price, under perfect competition price, MR and AR will be the same and the ***price line, MR curve and AR curve will be a horizontal straight line parallel to the x-axis.*** The demand curve (AR curve) facing a seller under perfect competition is shown below.



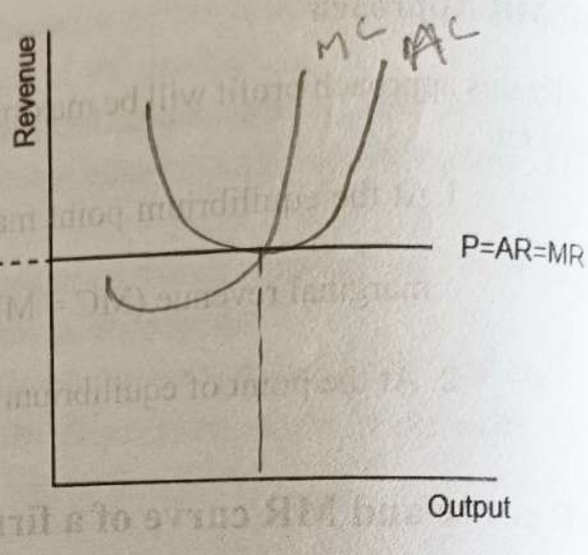


Fig. 5.2 Equilibrium of industry   Fig 5.3 AR, MR and price

In the above diagram the industry is in equilibrium at point E where market demand curve intersect market supply curve. P is the market price determined and this price is accepted by a firm. At this price P a firm can sell any amount of the commodity. Therefore the demand curve or AR curve is perfectly elastic.

Under any market condition AR curve is the demand curve

MC, MR Approach under perfect competition

A firm is in equilibrium when it gets maximum profit. Profit will be maximum when it satisfy the two equilibrium conditions. That is ***MC = MR and MC is rising at the point of equilibrium.*** MR curve of a firm is a horizontal straight line and MC curve is U shaped. Usually in the short period a firm may earn supernormal profit, normal profit or incur a loss. But in the long run a firm get normal profit only because whenever there is supernormal profit new firms will enter and when there is loss some of the existing firms will leave the market. In this way supply will be adjusted and the equilibrium price will increase or decrease accordingly.

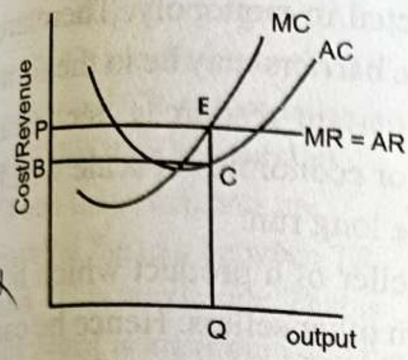


Fig 5.4 supernormal profit

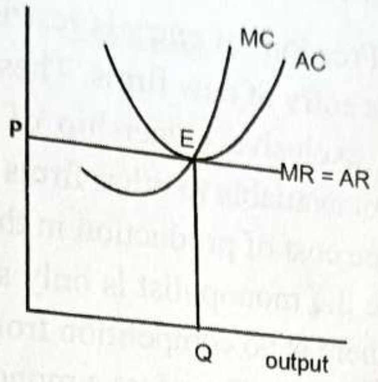


Fig 5.5 Normal profit

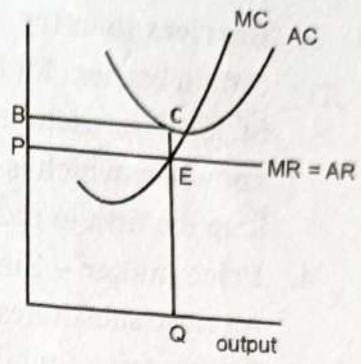


Fig 5.6 Loss

When there is supernormal profit at the point of equilibrium E, AC is less than AR. This gap EC is the profit per unit. The area PECB represents the total profit earned by the firm.

When the firm earn normal profit, at the point E, AR curve is tangent to the AC curve and hence AR and AC will be equal. Since AC includes normal profit, firm gets normal profit when it produce Q level of output. Mostly in the long run a firm get normal profit.

When there is a loss, at point E where MC equals MR, AC is greater than AR. The gap between AC and AR that is EC is the loss per unit and PECB is the total loss. When the firm produces Q level of output it can minimise the loss.

5.2 Monopoly

Monopoly means a single seller. ***It is a market situation in which a single seller controls the entire supply of a commodity***. In other words monopolist has the full control over price and output. Indian Railway is a monopoly of the government.

Features of monopoly

1. **Single seller** – under monopoly there is only a single seller who controls entire production and distribution of a commodity. Since there is only one seller there is no competition and the seller can charge any price for his product. Further, in monopoly there is no distinction between firm and industry because the firm itself is the industry.
2. **No close substitutes** – Monopolist is selling a product which has no close substitutes. Close substitute means goods which satisfy the same want.

3. **Barriers to entry** – Freedom of entry is restricted in monopoly. There are certain barriers for the entry of new firms. These barriers may be in the form of legal restrictions, exclusive ownership of certain resources, technical knowhow which is not available to other firms or economies of scale which help the firm to reduce cost of production in the long run.
4. **Price maker** – Since the monopolist is only seller of a product which has no close substitutes there is no competition from other sellers. Hence he can fix any price for his product. Therefore a monopolist is a price maker.
5. **Downward sloping demand curve** – Even though in monopoly price is fixed by the monopolist a consumer can decide whether he has to buy the product at that price or not. A consumer purchases a larger quantity only at a lesser price. That is a monopolist can sell a larger quantity at a lesser price and hence the demand curve or AR curve facing a monopolist is downward sloping.

MR curve and AR curve(demand curve) of a monopolist

As mentioned earlier a monopolist can sell a larger quantity only at a lesser price. Hence the MR curve of a monopolist will be downward sloping. As MR curve is downward sloping AR curve also will be downward sloping. But the AR curve will be less price elastic as there are no close substitutes in the market. Even if the monopolist changes the price there will not much change in demand.

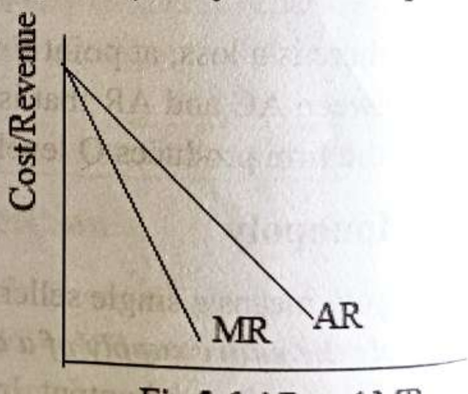


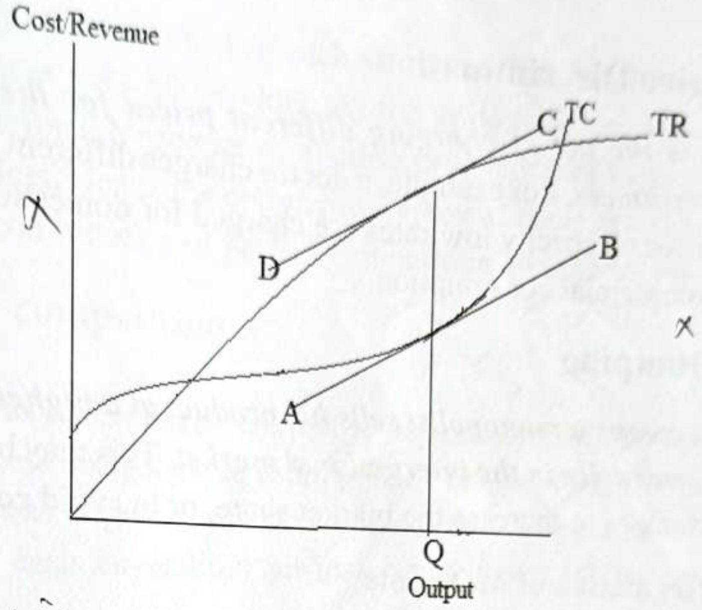
Fig.5.6 AR and MR

Equilibrium Price and Output Determination-

TC, TR Approach (Imperfect competition)

-A firm will be in equilibrium when he gets maximum profit. Profit is maximum when the difference between TR and TC is maximum. This situation is shown below.

In the diagram tangents AB and CD are drawn to the TC curve and TR curves in such a way that they are parallel. When these tangents are parallel the gap between TR and TC is maximum. That is the profit is maximum and the firm is in equilibrium and it produces Q level of output.



Equilibrium Price determination

Fig 5.8 TC, TR Approach

MC, MR Approach

Usually a monopolist earn supernormal profit because the monopolist has complete control over the supply of a commodity. Price and output determination is explained with the help of the following diagram.

In the diagram at point E, MC = MR and MC curve intersects MR curve from below. Hence E is the equilibrium point and OQ is the equilibrium level of output. When the firm produce OQ level of output QA is the AR or price. But the average cost is less than this and it is QB. Hence AB is the profit per unit. The rectangle PABC shows the total profit earned by the monopolist.

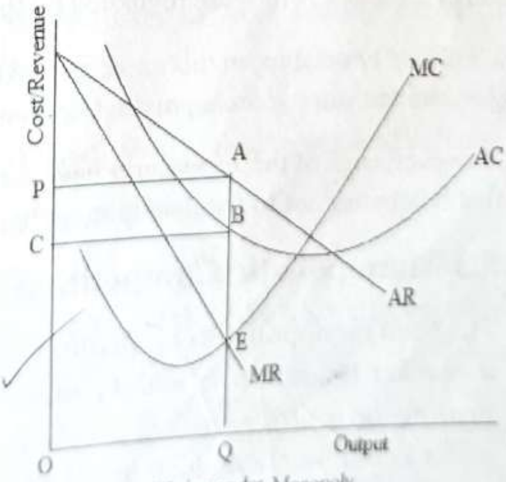


Fig 5 9 Equilibrium under Monopoly

However to control supernormal profit the government may impose a tax and this may reduce the profit. Hence there are rare chances for normal profit and loss to a monopoly firm. If there is loss the monopoly firm may closedown. (Diagrams are same as 5.12 and 5.13 to show normal profit or loss for a monopolist)

Price Discrimination

It is the act of charging different prices for the same product from different consumers. For example, a doctor charges different fees from poor and rich patients or for electricity low rates are charged for domestic consumption and high rate for commercial consumption.

Dumping

It means a monopolist sells his product at a higher price in the home market and lower price in the international market. This may be to clear the excess or outdated stock or to increase the market share, or to avoid competitors.

Regulation of Monopoly

Gregory Mankiw has suggested the following measures to control monopoly.

1. Increasing competition with Antitrust Laws: Antitrust laws are statutes developed by governments to protect consumers from monopoly practices and ensure fair competition. Antitrust laws allow the government to prevent merger. They also allow the government to breakup companies. Further, antitrust laws prevent the companies from coordinating their activities. In India there is the MRTP Act (Monopolies and Restrictive Trade Practices Act) to control monopolies.

2. Regulation: Through regulation the government does not allow the companies to charge any price as they wish. The government agencies regulate the price. For example, water and electricity charges are regulated by the government authorities.

3. Public Ownership: In this case, instead of regulating monopoly run by a private firm, the government run the monopoly itself. That is the government become the owner.

However, each of these measures have some drawbacks. Therefore, some economists argue that it is better not to regulate monopoly pricing.

5.3 Monopolistic Competition

The term monopolistic competition was given by Prof. Edward H Chamberlin. ***It is a market situation in which, there are a large number of buyers and sellers dealing in a differentiated product.*** Even though the product produced by each seller is not identical, they are close substitutes. They may differ in colour, shape, taste etc. Different brands of bath soap, soft drinks etc. are examples. Monopolistic competition is the real world situation.

Since the product is differentiated, the product of each seller has a unique feature. This give him a monopoly power over his product. At the same time product of each seller is a close substitute for the product of another producer and there are large number of buyers and sellers. Hence the competitive element is present. Thus monopolistic competition is a combination of perfect competition and monopoly.

Features of Monopolistic competition

1. **Large number of buyers and sellers** – Similar to perfect competition there are large number of buyers and sellers in monopolistic competition. But the number of sellers is not as large as in perfect competition. The number of buyers is so large that each buyer buys an insignificant part of the total output.
2. **Product differentiation** – Product differentiation is the essence of monopolistic competition. It can be in the form of changes in colour, shape, quality, packing etc. Therefore product of each producer has a unique feature and this give him the monopoly power over his product.
3. **Selling cost** – Each seller is selling a product which are close substitutes. Hence there is acute competition between sellers and they spend huge amounts on advertisement and other sales promotional activities. This is called selling cost.
4. **Freedom of entry and exit** – There are no restrictions on the entry or exit of firms. New firms can enter into the industry or loss making firms can leave the market at any time.
5. **Imperfect knowledge** – The information about market conditions like price, quality, cost etc. is not uniformly available to all buyers and sellers.

AR curve(Demand curve) and MR curve of a firm

AR curve or demand curve of a firm is downward sloping. This is because a seller can sell more only at a lesser price. Even though the seller has a small degree of monopoly power over his product, other sellers are also selling very close substitutes. Hence if he reduces the price he can get the customers of other sellers and hence the demand increases. On the other hand if he increases the price some of his customers will buy substitutes and hence the demand decreases. Thus demand curve is normal demand curve.

When compared to the demand curve of a monopolist it is moreflat or price elastic. In monopoly close substitutes are not available hence demand is less elastic. But in monopolistic competition products are close substitutes and hence a small change in price may cause a larger change in demand. MR curve lies below the AR curve.

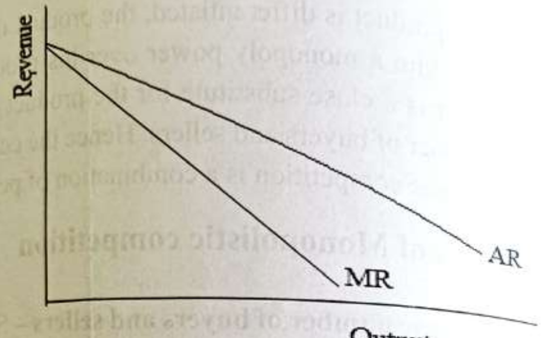


Fig 5.10 MR and AR

Price and output determination

A firm in monopolistic competition is in equilibrium when it maximises profit. Profit is maximum when it produces that level of output where MC=MR and MC is rising at the equilibrium point. Usually a firm earn supernormal profit in the short run. This situation is explained with the help of the following diagram. In the diagram at point E, MC = MR and at this point firm is producing OQ level of output. When production is OQ, average cost is QB but AR is greater than AC which is QA. Hence the difference between AC and AR, that is AB shows profit per unit of output. The rectangle PABC represent total profit earned by the firm.

Not all firms will earn Supernormal profit. Some firms may earn normal profit and some firms may incur loss. This is shown below.

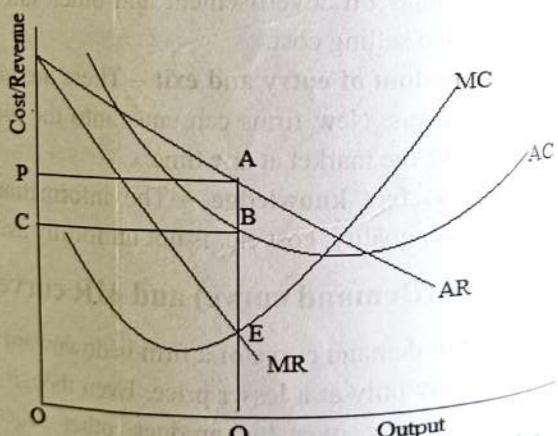


Fig 5.11 Equilibrium under Monopolistic competition

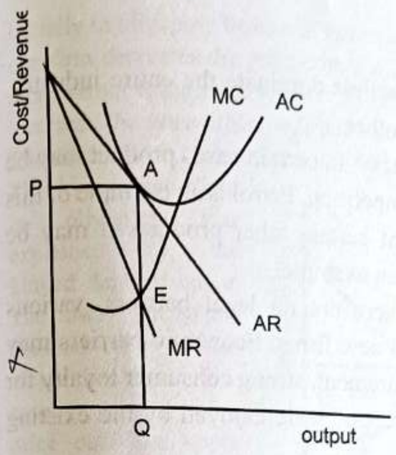


Fig 5.12 Normal profit

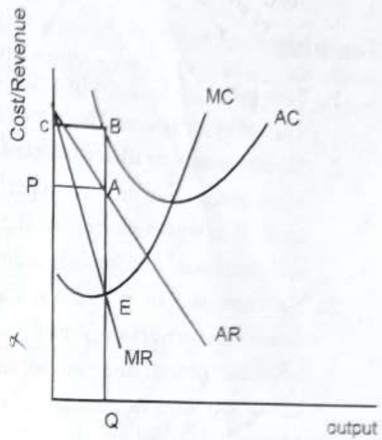


Fig 5.13 Loss

When a firm earn normal profit at MC = MR, AR curve will be tangent to the AC curve or AR=AC. This is shown fig. 5.12 where at point A, AR curve is tangent to AC curve. When the firm incur a loss AC curve lies above the AR curve or AC>AR. In the above diagram(5.13) rectangle PABC shows the loss of the firm. When the firm produces Q level of output where MC = MR, loss is minimised.

Freedom of entry and exit ensures **normal profit in the long run**. If there is high profit, new firms will enter and the market share(demand) of the firm decreases and the AR curve shifts downwards. Finally it becomes tangent to the AC curve. When there is loss some of the existing firms will leave the industry. Thus market share of the firm increases and therefore AR curve shifts upwards and it becomes tangent to the AC curve.

5.4 Oligopoly

Oligopoly is a market situation in which there are a few sellers selling either a homogenous or differentiated product. Aviation industry, telecommunication industry etc. are examples of oligopoly market forms. Since there are only a few firms they are interdependent to each other. Another important aspect of oligopoly is price rigidity.

eg cement indu motorbike

Features

1. **Few sellers** – Under oligopoly a few sellers dominate the entire industry. The sellers influence the price of each other.
2. **Homogenous or differentiated product** – In certain cases product may be homogenous like product in perfect competition. Petrol is an example of this kind of a situation. But in the case of certain other products it may be differentiated. Automobile industry is an example.
3. **Barriers to entry** – Even though there are no legal barriers, various economic barriers prevent the entry of new firms. Economic barriers may be in the form of huge investment requirement, strong consumer loyalty for existing brands or because of economies of scale enjoyed by the existing firms.
4. **Mutual Interdependence** –It implies that firms are influenced by each other's decision. The quantity sold by one firm depends on its own price and price of its competitors because the firms are producing either homogenous products or close substitutes.
5. **Existence of price rigidity** - It means that firms do not prefer to change the price of their product because it will not be beneficial for them. If one firm decreases the price others will also reduce price. Hence firm's product demand will not increase but at the same time it will affect their profitability. On the other hand if the firm increases the price others will not increase the price and hence it will lose its customers. Hence firms resort to non-price competition like advertisement and other forms of sales campaigning.
6. **Indeterminate Demand Curve** – This implies that the demand curve is unknown under oligopoly due to uncertain behaviour patterns of firms. Under oligopoly, every organization keeps an eye on the actions of rivals and makes strategies accordingly. Therefore, the demand under oligopoly is never stable and shifts in response to the actions of rivals.

Price and output determination-Kinked demand curve model

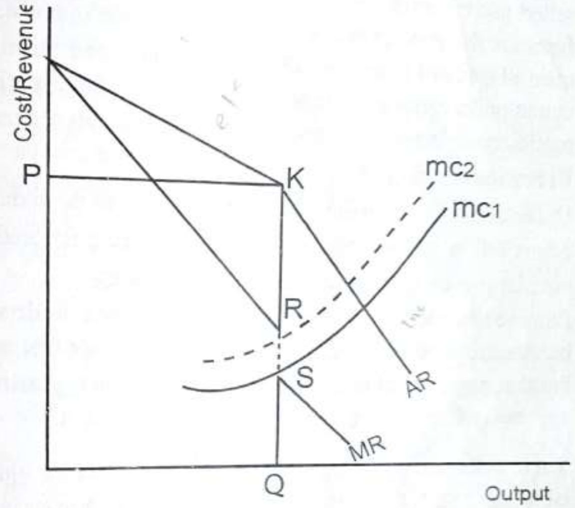
The kinked demand curve model was developed by Paul M Sweezy in 1939. Kinked demand curve explains price rigidity under oligopoly on the basis of following assumptions.

i. If a firm increases its price others will not follow.

ii. If a firm decreases its price others will also do the same.

Usually in oligopoly firms will not enter into a price war and hence price remains rigid. If one firm decreases the price others will also reduce price. Hence firm's demand will not increase but at the same time it will affect their profitability. On the other hand if the firm increases the price others will not increase the price and hence it will lose its customers. Therefore a firm under oligopoly stick to its price.

This kind of behaviour in oligopoly was explained by the kinked demand curve. The lower part of the demand curve is less elastic because a firm cannot gain from a price cut. The upper part of the demand curve is more elastic because there will be a substantial fall in demand if there is a price hike. Thus, in the following diagram we can see that there is kink at point K in the demand curve.



This kink in the demand curve or AR curve at point K creates a discontinuity in the MR curve. At the kink MR remain unchanged between points S and R.

Marginal cost curve mc1 intersect MR curve at point S and OQ is the equilibrium level of output. Suppose cost increases and MC curve shifts upwards as mc2 there will not be any change in equilibrium price and quantity till MC reaches the point R in the gap.

5.5 Collusive oligopoly

Under oligopoly firms are interdependent and face cut throat competition. ***To avoid price war and loss, firms enter into an agreement regarding uniform price and output. This agreement is known as collusion.*** According to Samuelson, "Collusion denotes a situation in which two or more firms jointly set their prices or output, divide the market among them, or make the business decisions".

Collusion helps the firms in preventing uncertainties, prevent the entry of new firms and strengthen the bargaining power of the firms against buyers. Collusion may be formal or tacit in nature. In formal collusion, there will be an explicit agreement among the firms. On the other hand, in tacit collusion firms collide in an informal way.

The most common form of explicit collusion is cartel. Cartel is an explicit or formal agreement between firms. This kind of collusion is possible when there are a few sellers and the product is homogenous. OPEC is an example of a cartel. Under cartel firms involve in price and output fixation, division of profit, market share etc. The main objective of cartel is reducing the supply and raising the price. Price and output under cartel are determined by a central administrative authority. The total profits are distributed in proportion as decided among the members.

Price leadership

Under collusion, sometimes the dominant firm in the industry sets the price and others follow it. The dominant firm becomes the price leader because of its large size, large economies of scale or better technology.

Barometric price leadership and aggressive price leadership also exist. Under barometric price leadership one firm change the price first and other firms follow it but that may not be the dominant firm. Like that in aggressive price leadership one aggressive firm sets the price first and others follow it.

Tacit collusion- Governments always take measures against the formation of cartels or formal agreements among the firms because such agreements are against the interest of the consumers. When firms enter into such agreements, they may act as a monopoly and charge a higher price. Therefore, firms inter into tacit collusion. In a tacit or implicit collusion firms do not form a cartel, but informally agree to charge the same price.

Non-Price Competition

Under oligopoly, there is very tight competition between the firms. If the firms try to increase their market share through price competition, it may result in a price war and hence the firms will be the losers. Hence, they resort to non-price competition to increase sales. Non-price competition refers to competition between companies that focuses on benefits, extra services, good workmanship, product quality etc.

Non-price competition is a marketing strategy that typically includes promotional expenditures such as sales staff, sales promotions, special orders, free gifts, coupons, and advertising. In other words, it means marketing a firm's brand and quality of products, rather than lowering prices.

There are two main branches of non-price competition. They are product differentiation and promotion or advertising. Product differentiation means differentiating the product

with respect to packing, colour, smell, quality etc. This helps to attract more customers and to increase the market share. Promotion includes advertising, branding, public relations etc. Advertising can be informative or persuasive.

The following are some of the examples of non-price competition.

Loyalty card – Loyalty cards give 'rewards or money back to customers who build up points. Airlines, supermarkets etc. use loyalty cards to encourage customers to repeat.

Subsidized delivery - Big firms such as Amazon has been successful in offering free delivery for their customers, with a paid subscription. This would give customers an incentive to purchase more because of the waived delivery fee. This works especially well for customers who are regular online shoppers. Supermarkets also offer delivery services for their customers.

Offering good after-sales service: After-sales service is crucial for the reputation and brand loyalty of the firm. In order to retain customers, they would have to provide great after-sales service.

Advertising/brand loyalty: Firms spend billions on advertising because repeated exposure to famous brands can make consumers more likely to buy such brands. High brand loyalty can also create barriers to entry.

Cultivation of good reviews: In an online world, good reviews are increasingly important. Therefore, firms have an incentive to encourage happy customers to leave reviews.

Coupons and free gifts: Some sellers provide coupons and free gifts along with a product. This encourages more customers to buy from that seller.

In short non-price competition increases the market share of a product. But it increases selling cost and other promotional expenses which in turn increases the average cost of production.

Comparison Between perfect competition, monopoly, monopolistic competition and oligopoly

Perfect Competition	Monopoly	Monopolistic Competition	Oligopoly
Large no. of buyers and sellers	Single Seller and large no. of buyers	Large number of buyers and sellers	Few sellers and large no of buyers
Homogenous product	Single product without close substitutes	Differentiated product	Homogenous or Differentiated
product			
Freedom of entry and Exit	Freedom of entry restricted	Freedom of entry and exit	Barriers to entry
No selling cost	No selling cost	Selling cost	Selling cost
Perfectly elastic demand curve	Downward sloping less elastic demand curve	Downward sloping more elastic demand curve	Indeterminate demand curve

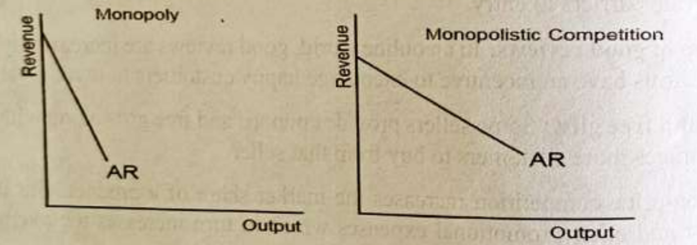


Fig 5.16 Demand curves under monopoly and monopolistic competition

5.6 Product Pricing

The right price of a product is one which keeps all participants of a market- consumers, sellers and shareholders- happy. A firm should decide its pricing strategy after considering the degree of competition in the market, price of competitors product, consumers buying capacity etc. Objective of the firm also play an important role in pricing decision because the objectives like profit maximisation and sales maximisation need different pricing strategies.

Another important factor which is to be considered in price fixation is cost of the product. This is the most important factor. Whenever there is a change in cost, the price of the product should be changed. Another factor which influences pricing policy is government policy regarding taxation and subsidy. Thus, it can be concluded that a change in anyone of the determinants of demand of a product necessitates a review of pricing policy. The following are the important pricing strategies.

Cost Plus or Markup Pricing

Under this strategy price is the sum of cost and a profit margin. Usually, average cost is used for this purpose. Therefore, it is also called Average Cost Pricing or Full Cost Pricing. Thus, the price will be

Price $= AC + m$ Where m is the percentage of markup. Markup is fixed arbitrarily and in many cases it is determined at 10 per cent. However, it may vary industry to industry and among the different firms in the same industry depending on the availability of substitutes, degree of competition etc. This method is very simple and convenient. However, one important limitation of this method is that it is not suitable when there is tough competition in the market or when there is the threat of entry of new firms.

Target Return Pricing

This method is similar to cost plus pricing. But the main difference is that cost plus pricing, the profit margin is decided arbitrarily, whereas under this method producer rationally decides the minimum rate of return. Even though the methodology of price determination is the same as the previous case, the margin is decided depending on the experience of the firm, consumer's paying capacity, risk involved and many other factors.

Penetration Pricing

When a firm wants to enter into a market which is already dominated by existing firms, the only option is charging a price less than the existing price. This price is called penetration price. Reliance has adopted this kind of a pricing strategy in the mobile phone industry. This method of pricing can be adopted on a short-term basis and its success largely depends on price elasticity of demand.

Predatory Pricing

Under predatory pricing the predator, already a dominant firm, sets its prices too low for a sufficient period of time so that its competitors leave the market and others are deterred from entering. This kind of predation is done on the expectation that these present losses (or foregone profits) will be compensated by future gains. In other words, the firm is on the expectation of acquiring exploitable market power after the predatory period, and that profits of this later period will be sufficiently large enough to compensate incurring present losses/c foregoing present profits. Predatory pricing usually will cause harm to the consumers and is considered as anti-competitive. It violates competition laws, as it makes markets more vulnerable to a monopoly.

Going Rate Pricing

This is the strategy of following the prevailing market price instead of a separate pricing strategy of their own. Usually, in this case price is fixed by a dominant firm and others accept it. Going rate pricing strategy is adopted when the products sold by the sellers are very close substitutes and their cross elasticity is very high. Packaged drinking water is an example. Besides, when new firms are not sure about the shift in demand in favour of them, they also follow this pricing strategy. By adopting this pricing strategy firms can avoid a price war like situation.

This kind of a pricing strategy is adopted when the product has reached maturity and has become generic in nature. That is a buyer ask for a product in general rather than a particular brand. An example is mineral water.

Price Skimming

It is a strategy in which high price is charged at the time of introduction of the product and a lower price during maturity. By experience producers know that a segment of high-income consumers wishes to become the first among those who possess the product. They use the product as a status symbol instead of considering its intrinsic value. Hence, the producers charge a very high price from such buyers to skim the market and earn a very high profit. Once the product is established and reached maturity, producers will reduce the profit margin and charge a lower price. This will attract the lower income group.

Administered Pricing

Generally, the term administered pricing is used to denote the price charged by the monopolists. Since, a monopolist is a price maker he can charge any price for his product. In other words, administered prices are not fixed by the market mechanism.

But in the Indian context, administered pricing means price is fixed statutorily by the government. During certain occasions government fixes the price of certain essential commodities on social interest. Price of cooking gas is an example.

Chapter 6

National Income

6.1 The Circular flow of Income

Economic transactions generate two types of flows i) Product flow or real flow and ii) money flow. In the economy products and money flow in opposite directions in a circular manner. This is called circular flow of income. Product flow includes flow of goods and services, and factor services.

When factors of production supply their factor services, they get factor incomes. This is the income flow. This income is spend on various goods and services and creates expenditure flow. To illustrate the flows of income and expenditure, an economy is divided into four sectors- i) Household sector ii) Business sector or firms iii) government sector iv) Foreign sector.

Circular flow in a two sector model

In a simple two sector model the two sectors are households and firms. Households possess all factors of production. They supply these factor services to firms and get factor payments in the form of rent, interest, wages and profit. This income is spend for buying goods and services produced by firms.

Firms hire the factor services of households and produce various goods and services. They sell these goods and services to the households. These flows are depicted in the following chart. This model is built on the basis of the assumption that the entire income received by the households are spent on goods and services. That is,

$Y \equiv FP \equiv V$ Where Y=household income, FP=factor payments, V=money value of output

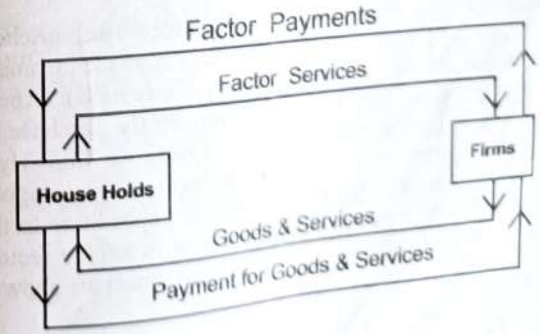


Fig 6.1 Circular flow in a simple two sector model

In the above diagram, upper part represents the factor market and the lower part represents commodity market. In the factor market there is a flow of factor services from the household sector to the firms. In return there is a flow of factor payments from firms to households. In the commodity market there is a flow of goods and services from the firms to household sector. In return there is a flow of payment for goods and services from household sector to the firms. Thus the money flows and real flows are completed.

Two sector model with capital market

Households may not spend their entire income on goods and services. They may save a part of their income. Unspent income is a leakage from the economy. This saving is coming to the capital market. Firms borrow the savings of the households and they invest. Investment is an additional expenditure and it is an injection. When saving equals investment there will not be any problem in the economy. Two sector model with capital market is shown below.

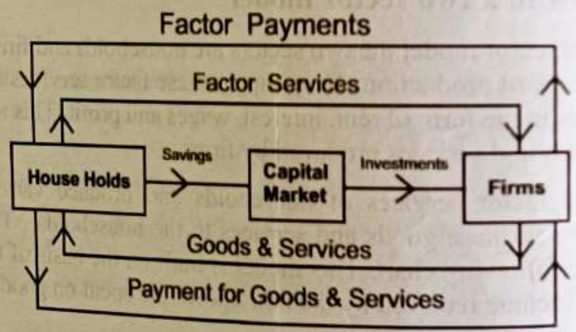


Fig. 6.2 Two sector model with capital market

Three sector model

In a three sector model the government sector is included. The household sector and the firms pay tax to the government. Tax is a leakage from the income stream. Like other two sectors government also spend money. Government expenditure is an injection. Government make payment to the firms for the purchase of goods and services. Further government also pay subsidies to the firms. Similarly, government make use of the manpower services of the households and pay wages and salaries as well as transfer payments in the form of welfare expenditure to the household sector. The following figure shows the circular flow in a three sector model. For convenience only money flows from and to the government are shown.

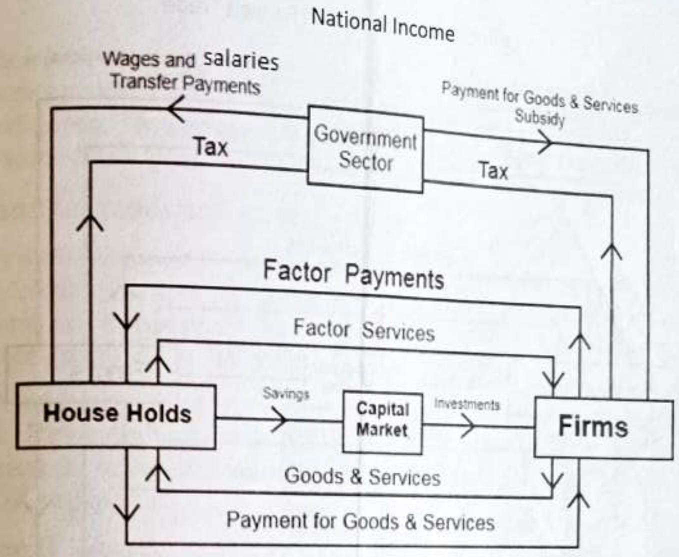


Fig 6.3 Circular flow in a three sector model

If the government expenditure equals taxes then there is no disturbances to the circular flow. This is the case of balanced budget. On the other hand if government expenditure is less than the taxes, then there is leakage and the size of the circular flow will be reduced. This is the case of surplus budget. But, when government expenditure is greater than taxes there is an injection into the economy and the circular flow will be expanded. This is the case of deficit budget.

Circular flow in a four sector model

In a four sector model the fourth sector is the foreign sector. Households export their manpower to the foreign sector and in return they get foreign remittances. Firms export their goods and services to the foreign sector and they get receipts from exports. At the same time firms import raw materials, and other inputs from the foreign sector and they make payment for their imports. Export is an injection into the economy but import is a leakage. The circular flow in a four sector model is shown in the following figure.

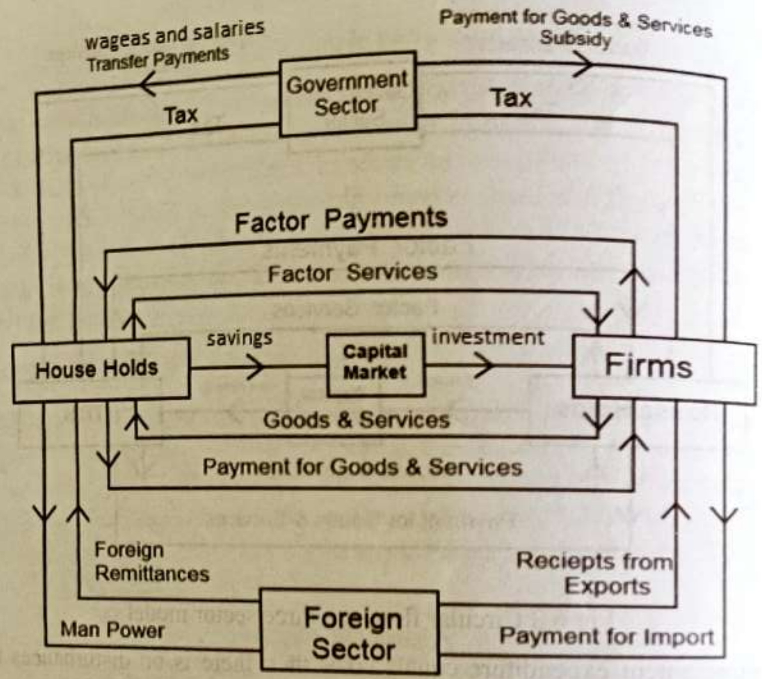


Fig. 6.4 Circular flow in a four sector model

6.2 National Income

Factor income and Transfer income

Factor income is the income received for supplying a factor service. It can be in the form of rent, interest, wages or profit. When a factor payment is made either a product or service is produced. For estimating national income factor payments alone are taken into account.

On the other hand ***transfer payments are unilateral or one sided payments***. They do not add anything to the current flow of goods and services. Donations to charity, unemployment allowance, old age pension, gifts etc. are examples of transfer payments. Tax is a compulsory transfer payment. Transfer payments are excluded from national income accounting.

Intermediate goods and final goods

Goods which are used in the production of other goods and services are called intermediate goods. They pass through further stages of production. Raw materials, fuel, electricity etc. are intermediate goods. Intermediate goods are not taken into account for national income estimation.

Goods which are ready for consumption or investment are called final goods. Consumer goods like dress, vehicles and electronics items, machinery etc. examples of final goods. They do not need further processing. They are finished products. The value of final goods are added for the estimation of national income.

Consumer goods and capital goods

Goods which are used for consumption purpose are called consumer goods. Food items, clothing, household electronics and electrical items etc. are examples of consumer goods. Consumer goods can be durable like TV, fridge etc., semi durable like dress, shoes etc. or none durable like food items.

Goods which are used to produce other goods and services are called capital goods. They are used for investment purpose. Machinery, equipment, building etc. are examples of capital goods. The economic growth of a country depends on the stock of capital goods.

National Income

National income can be defined differently. From the income side ***it is the sum total of the factor incomes received by the residents of a country in the form of rent, interest, wages and profit over a period of one year.*** From the output side it is the total money value of all final goods and services produced in an economy during an accounting year.

Gross Domestic Product at Market Price(GDP_{mp})

It is the money value of all final goods and services produced within the domestic territory of a country during a financial year. Money value is the price of the product. To estimate GDP money value of final goods alone are taken into account. Further the value of goods and services produced in the domestic territory alone will be taken into account. That is goods produced outside the country by its nationals will not be considered.

Net Domestic product at market price(NDP_{mp})

When depreciation is deducted from GDP we get NDP. Depreciation is the loss in the value of capital assets due to wear and tear during production. That is

NDPmp = GDPmp – Depreciation. Therefore to find net value from any given gross value depreciation is to be deducted.

Net National Product at market price (NNPmp)

GDP includes the money value of goods or income generated within the domestic territory only. But a nation also get income from abroad. When net factor income from abroad (NFIA) is added to NDP we get NNP. Thus the difference between domestic product and national product is NFIA. NFIA is the difference between factor incomes received from abroad and factor payment made to the rest of the world. For example every year India get factor income in the form of rent, interest, wages and profit from other countries. At the same time factor payments are made to other countries for making use of their factor services. The difference between these two is called NFIA.

NNPmp = NDPmp + NFIA

Net National Product at factor cost (NNPfc)

imp

Market price includes indirect taxes and subsidies. When the effects of these two are eliminated from the value of output we get the value of output at factor cost. To estimate the value at factor cost net indirect tax is to be deducted. Net indirect tax is the difference between indirect tax and subsidy.

NNPfc = NNPmp – Net Indirect Tax(NIT)

NIT = Indirect Tax – Subsidy

NNPfc is the national income of a country.

Gross National Product (GNP)

GNP is the money value of all final goods and services produced in a country including net factor income from abroad.

GNP = GDP + NFIA

Numerical example

From the data given below estimate GNPmp, GNPfc, NNPmp and National income.

GDPmp= 5000 (in 100 billion)

NFIA = -50

Indirect Tax = 70

Subsidy = 20

Depreciation =30

Answer

GNPmp= GDPmp+NFIA

= 5000 + -50 = 4950

GNPfc =GNPmp – NIT

= 4950 – (70 – 20) = 4900

GNPmp − 70 + 20 (In, Sub)
= 4900

NNPmp= GNPmp – Depreciation (or GDPmp –Depreciation + NFIA)

= 4950 – 30 = 4920

National Income (NNPfc) = NNPmp – NIT

= 4920 – (70 – 20) = 4870

Private Income

Private Income refers to income of non-governmental entities from all sources over a period of one accounting year. It represents income of firms and households from all possible sources.

Private Income = NNP_{FC} – domestic product accruing to the government sector + transfer payments + Interest on public debt

Personal Income

It is the income of the household sector from all sources before paying direct taxes in a financial year.

Personal income (PI) ≡ National Income – (Corporate tax + Undistributed corporate profits + social security contributions) + Transfer payments + interest on public debt

(Corporate tax is tax on profit of the companies)

Personal Disposable Income

It is defined as the part of personal income left for consumption and saving after the payment of taxes.

Personal Disposable Income = Personal Income – Direct taxes

Per capita income

It is the income per head. In other words it is the average income of the people of a country in one year. It is obtained by dividing national income by population.

Per capita income = $\frac{\text{National Income}}{\text{Population}}$

National Income at Current Prices and National Income at Constant Prices

In an economy without any increase in real output national income estimate may show an increase in national income. This is because of the increase in price of goods and services. Hence in every country national income is estimated at current prices and constant prices.

National income estimated according to the prices of goods and services prevailed in the current year(the year to which national income is estimated) is known as national income at current prices or nominal national income. From this we will not get a clear picture of the economy. Therefore national income at constant prices is estimated.

National income estimated according to the prices of goods and services prevailed in the base year is known as national income at constant prices or real national income. When there is an increase in national income at constant prices it shows that there is economic growth in that country or the output has been increased. In India 2004-05 was taken as the base year for estimating national income at constant prices. But recently the base year has been changed to 2011-12. That is in India in every year national income will be estimated according to current year price and on the basis of prices of goods and services prevailed in the year 2011-12.

The GNP Deflator

This is an adjustment factor used to convert nominal GNP into real GNP. It is the ratio of price index number (PIN) of a chosen year to the price index number(PIN) of the base year. PIN of the base year is 100. The chosen year is the year whose real GNP is to be estimated.

> PIN is a percentage number that shows the average change in the price of a basket of goods over a period of time as compared with prices in the base year. Base year is a standard year taken for comparison and its PIN is taken as 100. If PIN for a year is 125 it means that there is 25% increase in the price of goods and services when compared to the base year.

$$\text{GNP Deflator} = \frac{\text{PIN of the chosen year}}{100}$$

$$\text{Real GNP} = \frac{\text{Nominal GNP}}{\text{GNP deflator}}$$

The Three Sectors of an Economy

Economic activities are classified under three sectors. They are primary, secondary and tertiary sectors.

Primary sector consists of activities related to the exploitation of natural resources. The main activities under primary sector are agriculture, mining and quarrying, forestry, fishing, animal husbandry, poultry farming etc.

Secondary sector is the manufacturing sector. Secondary sector includes registered and unregistered manufacturing.

Tertiary sector provides various services like health, education, banking, insurance, transport and communication, trade and commerce, hotels and restaurants etc.

In developed countries the tertiary sector contributes the largest share towards national income. Even though India is a developing country the largest share of its national income is contributed by the tertiary sector.

Stock and Flow

Stock is the quantity of a variable measured at a point of time. It has no time dimension. It is a static concept. Water in reservoir at a point of time is a stock. In economics wealth, capital etc. are stocks.

Flow is the quantity of a variable measured over a period of time. It has a time dimension, may be a week, month, year etc. It is a dynamic concept. Water accumulated in the reservoir between a period of time is flow. Similarly cash in hand is stock but monthly income is flow. Saving, expenditure etc. are also examples of flow.

6.3 Measurement of National Income

There are three important methods of measuring national income. They are

1. Product method or output method
2. Final Expenditure method
3. Income method

Product method or Output method

Under this method GDP is estimated as the sum of the money value of all final goods and services produced in the domestic territory of a country during a financial year. We can arrive at national income once GDP is estimated. The following are the important steps involved in the estimation of GDP

i) Identifying the production units and classifying them under respective industries and each industry under the corresponding sector.

ii) Estimate the value of final output produced by each production unit, each industry and each sector.(Gross value of output of a production unit = P*Q where P is the price per unit and Q is the number of units of output produced in a year)

The sum of value of output produced by all the three sectors gives GDPmp. That is

$\Sigma GVOmp = GDPmp$

Once GDPmp is estimated we can derive NNPfc by deducting depreciation and net indirect tax and adding NFIA. NNPfc is the national income.

If total sales during a financial year is given along with opening stock of goods and closing stock, then

GVOmp = Total sales + Change in stock

Change in stock = Closing stock – opening stock

While using this method the value of goods produced for self consumption is also added.

However this method has the problem of double counting. Double counting means counting the value of a product more than once. This difficulty arise because final product of one firm becomes the intermediate product of another producer. Hence it is difficult to classify the goods as final product and intermediate product. Double counting leads to overestimation of national income. This problem can be solved by using the value added method.

Under the value added method instead of taking the value of output the gross valued added by each production unit is estimated. Gross value added is the difference between the Gross value of output and intermediate consumption. Intermediate consumption means the value of intermediate goods used in the production of a commodity.

Gross Value Added at market price(GVAmp)= GVOmp - Intermediate Consumption

Net Value Added at market price = GVAmp - Depreciation

ΣGVAmp = GDPmp

Suppose a baker produces bread for Rs.1000 by using the inputs like milk, wheat flour, sugar etc. for Rs.600. Value added by baker is 1000 – 600 = 400. If we add the value of bread as well as milk, sugar and wheat flour in national income estimation, double counting arises.

Final Expenditure Method

This method estimate GDP by adding the final expenditures in the economy. There are four major components of final expenditure

i)Private final consumption expenditure(C) – This is mainly the household expenditure on final goods and services to satisfy their wants. It mainly depends on their disposable income.

ii) Investment Expenditure (I) – This is the expenditure for acquiring capital assets. There are four major components investment expenditure a) Private investment by firms b) Government investment or autonomous investment c)

Residential construction investment d) Inventory investment (Unsold stock of goods)

iii) Government consumption expenditure (G)–Like households, government also spend money for purchase of consumer goods like stationery, petrol etc.

iv) Net exports (X-M) – This is foreign expenditure. Foreigners spend money for our goods(export) and we purchase foreign goods(import). The difference between these two is the net foreign expenditure.

When these four items are added we get GDPmp. That is

C+I+G+X-M = GDPmp

Once GDPmp is estimated we find NNPfc or national income.

Income Method

When production takes place income is generated in the form of factor payment. That is when output worth Rs. 100 is produced income equal to rupees 100 is generated in the economy as factor payments. Income method take the sum of the factor incomes in the economy. Factor incomes are

i) Rent(R) – It is the income earned by the people who supply land and building in production.

ii) Interest(I) – It is the reward of capital. When money is borrowed for investment interest is paid as the reward.

iii) Wages (W)– It is the reward of those who supply labour power. It is also called compensation of employees.

iv) Profit (P)– Profit is the reward of entrepreneur. Entrepreneur is the person who combines the services of other factors of production, take risk and produce various goods and services.

Even though these are the four factor incomes one more item is added. That is the income of the self employed. The income of self employed people is a mixture of rent, interest, wages and profit and it cannot be separated. Therefore the fifth item is added as

v) Mixed income of the self employed.

When these five items are added we get NDPfc.

R+I+W+P+ Mixed income = NDPfc (If Gross profit is added then we get GDPfc because Gross profit includes depreciation also)

NNPfc = NDPfc + NFIA

Since we are adding factor incomes net indirect taxe is not taken into account. When NFIA is added to NDPfc we get national income. ***The sum of Rent, interest and profit is called operating surplus.***

Items excluded from national income estimation

The following transactions are not included in national income estimation

1. Buying and selling of shares and securities. These are only financial transaction. Nothing is produced because of these transactions.
2. Value of intermediate goods used.
3. Prize money from lottery.
4. All transfer payments.
5. Purchase and sale of second hand goods (value of these goods are included when it was purchased for the first time).
6. Income from illegal activities like smuggling, gambling etc.

Uses or significance of national income estimation

- To evaluate the performance of the economy over the years
- For economic planning and for the formulation of economic policies
- To understand the contribution of each sector towards national income
- To make comparison between the economic performance of two countries
- To measure the inequalities in the distribution of income

Difficulties in the measurement of national income

There are two types of difficulties in the measurement of national income. They are conceptual difficulties and statistical or practical difficulties. Conceptual difficulties are common to developed as well as developing countries. But practical difficulties are mainly applicable to developing countries.

Conceptual difficulties

1. Service without remuneration- Certain services such as service rendered by a house wife is not included in national income because payments are not made. But, when the same services are supplied by a house-maid remuneration is paid and it is included in national income estimation.

2.Classification of goods as intermediate goods and final goods – It is very difficult to classify certain goods as final goods and intermediate goods because the

same product is used as an intermediate good and final good. When milk is purchased by a household it is a final good but in a hotel it is an intermediate good.

3. Difficulty in estimating the value of output produced in the government sector.

Since the government provide public goods either at free of cost or at nominal prices it is very difficult to estimate their value.

Practical difficulties

1. Inadequacy of statistical data. – In developing countries accurate and adequate statistical data is not maintained.

2. Illiteracy of farmers- Farming is the main activity in developing countries but majority of the farmers are illiterate. Hence they do not keep proper accounts of their production.

3. Lack of occupational specialisation – In developing countries people are mostly unskilled and they earn their income from more than one occupation. Hence it is difficult to compute their income.

4. Production for self consumption- In developing countries a major part of the agricultural output is consumed by the farmers themselves. Hence its value cannot be estimated.

5. Existence of a non monetised sector – In developing countries, in villages people still make certain barter transactions.

Besides, black money, price changes etc. also pose problems to national income estimation.

Numerical example

From the data given below estimate national income according to value added method, expenditure method and income method.

	(Rs. Crores)
Gross Value of Output at market price	8000
Intermediate consumption	2000
Private consumption expenditure	3000
Investment expenditure	2000
Government expenditure	700
Exports	600

Imports	300
Wages and salaries	2000
Rent	500
Interest	500
Profit	1500
Depreciation	1000
Indirect tax	800
Subsidy	300
Net factor income from abroad	-500

Value Added Method

GVAmp = GVOmp-Intermediate consumption

= 8000-2000 = 6000

Therefore GDPmp = 6000

NDPmp = GDPmp – Depreciation

= 6000-1000 = 5000

NNPmp = NDPmp+NFIA

= 5000+(-500) = 4500

NNPfc = NNPmp – NIT

= 4500-(800-300) = 4000

National Income = Rs. 4000 Crores

Expenditure method

GDPmp = C+I+G+X-M

= 3000+2000+700+(600-300) = 6000

NDPmp = GDPmp – Depreciation

= 6000-1000 = 5000

NNPmp = NDPmp+NFIA

= 5000+(-500) = 4500

6000 - intermediate consumption

NNPfc = NNPmp – NIT

= 4500-(800-300) = 4000

National Income = Rs. 4000 Crores

Income method

NDPfc = W+R+I+P = 4500

= 2000+500+500+1500 = 4500

NNPfc = NDPfc+NFIA

= 4500+(-500) = 4000

National income = Rs. 4000 crores

2) A bicycle manufacturing company in India produced and sold 100 bicycles at a price of Rs.2500 per unit in the market. Out of this Rs.300 has gone to the government as tax per unit. The owner of this company is a foreigner and he got a profit of Rs.500 per bicycle and the entire profit has gone to the country to which he belongs. Because of the production of 100 bicycles there was a depreciation of Rs.20,000 to the company. How much is the contribution of this company to GDPmp as well as national income of India?

Gross value of output at market price = Price * Quantity = 2500*100 = 2,50,000

Therefore contribution to GDPmp = Rs.2,50,000/-

Contribution to national income= Contribution to GDPmp - Depreciation + NFIA-NIT

(As Profit is going out of the country contribution to NFIA is negative)

= 250000 - 20000 +(-500*100) - 300*100

= Rs.1,50,000/-

Chapter 7

Inflation

7.2 Inflation- Meaning and Types

Inflation is a situation in which there is a persistent rise in the general price level. In other words it is a situation in which there is an upward movement in the average level of prices. According to Coulborn it is a situation in which "too much money chasing too few goods. That is the availability of goods is less when compared to the supply of money. When there is inflation value of money decreases persistently. Value of money is the purchasing power of money or the quantity of goods and services that a unit of money can purchase.

There are several types of inflation which are classified on different basis. Based on the rate, inflation can be classified as Creeping, Walking, Running and Galloping Inflation.

(a) **Creeping Inflation**: When the rise in prices is very slow, that is less than 3% per annum, it is called creeping inflation. It is mild inflation and it is considered as good for economic growth.

(b)**Walking Inflation:** When prices rise moderately and the annual inflation rate is 3% to 10%, it is called walking inflation. Inflation at this rate is a warning signal for the government.

(c) **Running Inflation:** When prices rise rapidly and the rate of increase is 10% to 20% per annum, it is called running inflation. Its control requires strong monetary and fiscal measures and it is a dangerous situation.

(d) **Galloping or Hyperinflation:** When price rises between 20% to 100% per annum or even more, it is called galloping or hyperinflation. Such a situation brings a total collapse of the monetary system because of the continuous fall in the purchasing power of money.

Demand Pull Inflation and Cost Push Inflation

Demand pull inflation is the result of an increase in aggregate demand in the absence of an increase in aggregate supply or a relatively less increase in aggregate supply. Suppose the government is following an expansionary fiscal policy and the government spend more money in the field of education, health etc. by printing more currency notes. In this situation there will be a sudden increase in aggregate demand in the economy but the aggregate supply will not increase to that extent. Therefor the price level goes up. Once the economy reaches in full employment

level any further increase in aggregate demand will lead to price rise without any increase in output. This can be explained with the help of the following diagram.

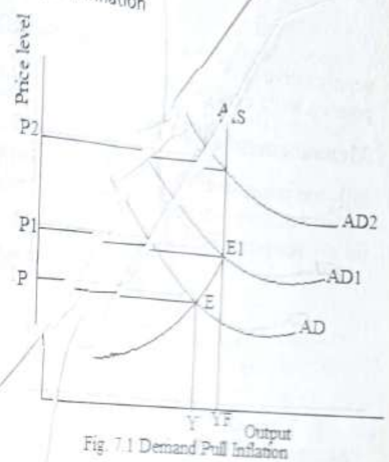


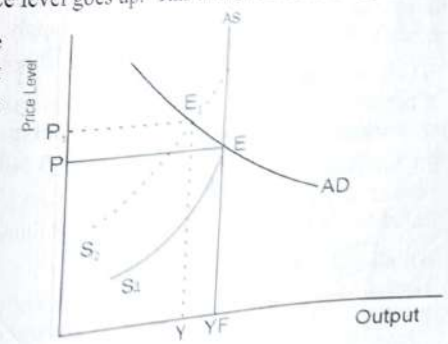
Fig. 7.1 Demand Pull Inflation

In the diagram, AD is the aggregate demand curve, AS is the aggregate supply curve. Initially the economy is in equilibrium at point E. Y is the equilibrium level of output and P is the price level. When aggregate demand increases AD curve shifts upwards and the new Ad curve is AD1 which intersects the AS curve at point E1. Here price level increases to P1 and there is an increase in output from Y to Y_F. Beyond the Y_F level of output AS curve becomes perfectly inelastic. That is the output cannot be increased beyond this level of output. Hence YF is the full employment level of output. Any increase in aggregate demand beyond this level will push the price up without any change in output. The theory of demand-pull inflation is associated with the name of J M Keynes.

Cost push inflation is the result of increase in cost of production. Cost of production increases mainly due to increase in wages, increase in profit margin, or due to a supply shock which means a sudden fall in supply. Increase in cost of production decreases the supply and when supply decreases, supply curve shifts leftwards. Therefore the price level goes up. This is shown in the f diagram.

In the diagram initially the economy is in equilibrium at point E where the aggregate supply curve AS1

Fig.7.2 cost push inflation



intersects the AD curve. This is full employment equilibrium where output is Y_F and P is the price level. When aggregate supply decreases the AS curve shifts leftwards and the new supply curve is AS 2 which intersects the AD curve at E1. Therefore the price level goes up to P1 and output decreases to Y.

Measurement of inflation

Inflation is measured by calculating the changes to price index numbers (PINs) over a period of time. Rate of inflation is the percentage rate of change in the price index for a given period of time.

$$\text{Rate of inflation} = \frac{PIN_t - PIN_{t-1}}{PIN_{t-1}} * 100$$

Where PIN_t= Price index number for year t

PIN_{t-1} = Price index number for year t-1

Price index measures the average change in the price of goods and services over a period of time. It can be consumer price index or whole sale price index.

Causes of Inflation

Causes of inflation can be classified under demand side causes and supply side causes.

Demand side causes

In an economy when aggregate demand increases without an equivalent increase in aggregate supply, there will be excess demand and the price level goes up. This leads to demand pull inflation. The following are the important causes of demand pull inflation.

i) **Increase in money supply**–This is the most important reason for inflation. When the monetary authority increases the money supply, cash in hand with the people increases and hence they spend more money. Thus aggregate demand increases.

ii) **Increase in disposable income-** Disposable income increases due to an increase in per capita income or reduction in taxes. Increase in disposable income also increases cash in hand with the people and aggregate spending.

iii) **Increase in government expenditure** – When the government follow an expansionary fiscal policy government expenditure will increase and as a result demand for goods and services also increase.

iv)**Deficit financing** – It means government spends more money than its revenue. The deficit may be met by printing more currency notes. This also increases money supply as well as total spending in the economy.

v) **Cheap money Policy** – If the interest rates are low in the economy incentives to save will be less. Hence people spend more money and thus aggregate demand increases.

vi) **Increase in population** – When population increases the number of buyers also increases and thus aggregate demand increases.

Supply side causes

When aggregate demand increases and aggregate supply does not keep up with aggregate demand, cost push inflation will be the result. There are different reasons for inadequate aggregate supply.

i) **Shortage of capital and other complementary factors** – To increase production and aggregate supply more capital and other complementary factors like raw materials, electricity etc. are needed. When there is a shortage of such factors there will be less aggregate supply and as a result price level goes up.

ii) **Increase in wages** – When wage rate increases cost of production also increases. As a result supply falls and price level goes up.

iii) **Speculative hoarding** – When traders hoard goods and create artificial scarcity to get more profit, price will increase.

iv)**Natural calamities** – Natural calamities like drought, earthquake etc. reduces production and aggregate supply.

v) **Increase in exports** – When exports increases availability of goods in the domestic market decreases.

vi) **Industrial disputes** – Industrial disputes will affect industrial production and aggregate supply.

Effects of Inflation

Effects of inflation can be studied under

1. Effects on distribution of income 2. Effects on investment and production 3. Social and political effects.

1. Effects on distribution of income and wealth- The effects of inflation on distribution of income and wealth can be viewed as the effects on fixed income group and flexible income group. When there is inflation the poor and the middle class whose income is relatively fixed will lose but the flexible income group categories businessmen, industrialists, traders, real estate holders and speculators gain and the income distribution change in favour of them. These categories are analysed one by one.

a) Debtors and Creditors- During inflation, debtors gain and creditors lose. Because of inflation value of money decreases and therefore people who lend their money, when they get it back its value will be less and they can purchase only less amount of goods and services.

b) Salaried classes and wage earners – Since the income of these two groups will adjust to inflation very slowly they will lose when there is inflation.

c) Investors – Those who invest in shares will gain because companies will make more profit when there is inflation. On the other hand those who invest in bonds and debentures which carry fixed returns will lose.

d) Businessmen – Since price goes up business people get more profit and they gain from inflation.

e) Farmers – When there is inflation price of agricultural product increases at a faster rate. Hence farmers get more income.

2. Effects on investment and production –When there is inflation people will have tendency to spend more and save less. Hence there will be less savings and less investment in the economy which will adversely affect production. Inflation also discourages foreign investment.

However in the initial stages, producers and traders get more profit because of price rise. Hence they produce their maximum and thus production goes up. But later, inflation increases wage rate and price of raw materials. Hence cost of production increases and as a result output may decrease.

Besides inflation may lead to misallocation of resources because producers will divert their resources from the production of essential commodities to those goods which gives them maximum profit. Another adverse impact of inflation is black marketing. Traders may hoard stock of goods to create artificial scarcity to make more profit by selling it at a higher price.

3. Social and Political impact - Inflation makes the rich richer and the poor poorer. Hence people will be unhappy. Because of the rising cost of living, workers resort to strikes which lead to loss in production. To make more profits, people resort to hoarding, black marketing, adulteration, manufacture of substandard commodities, speculation, etc. Corruption spreads in every walk of life. All this reduces the efficiency of the economy. Thus there will be social unrest in the economy.

If hyperinflation persists and the value of money continues to fall, it ultimately leads to the collapse of the monetary system. Further, rising prices also encourage agitations and protests by political parties opposed to the government. This may lead to the downfall of the government.

Measures to control inflation

There are three important ways in which inflation can be controlled.

1. Monetary policy measures
2. Fiscal policy measures 3. Other measures

1. Monetary policy measures

These are the measures adopted by the central bank of a country to control credit and money supply in an economy. Price stability and economic growth are the two main objectives of monetary policy. Monetary policy measures can be classifies as a) Quantitative credit control measures b) Selective or qualitative credit control measures

Quantitative credit control measures

Quantitative controls aim at regulating the overall volume of bank credit, without considering purpose for which credit is used. The important quantitative measures are

i) Bank Rate Policy – ***The Bank rate** is the rate at which the central Bank **rediscount approved bills of exchange.*** According to RBI, it is the rate at which bills of exchanges and commercial papers are rediscounted or bought. During inflation, the central bank raises the bank rate due to which the cost of borrowing goes up. As a result, commercial banks borrow less money from the central bank. With the reduced borrowings from the central bank, the flow of money from the commercial bank to the public decreases.

Further, as the central bank raises the interest rate, the commercial banks also raise their lending rate to the public, thereby making the borrowings costlier. Hence people take less loans from the commercial banks and spend less money. This helps to reduce money supply as well as aggregate demand in the economy.

Higher rate of interest creates an adverse environment for business activities and hence business activities contracted. Similarly, higher rate of interest is an incentive to save more and spend less. All such changes in the economy because of a higher rate of interest reduces the total spending as well as aggregate demand in the economy.

Increase in bank rate is called dear money policy and decrease in bank rate is called cheap money policy. When bank rate is increased cost of borrowing increases and hence money becomes dearer and when it is decreased money becomes cheaper.

ii) Reserve Ratio–Depending upon the economic conditions central bank increases or decreases the reserves that every commercial bank should keep in the central bank. There are two types of reserve ratios:

Cash Reserve Ratio(CRR)-***Every commercial bank should keep a certain percentage of their total deposits(net demand and time deposits) in the central bank in the form of cash reserve. This is mandatory and this percentage is called CRR***. According to RBI it is the average daily balance that a commercial bank is required to maintain with the RBI as a percentage of its total deposits. When there is inflation the central bank increases the CRR. This reduces the availability of cash with the commercial banks and their lending capacity decreases. Hence people get less money in the form of loans from the commercial banks and it helps to control inflation.

Statutory liquidity ratio (SLR) – This is another type of reserve that every commercial bank should keep. ***A commercial bank should keep a certain percentage of their total deposits in the form of safe and liquid assets such as unencumbered government securities, cash and gold.*** When there is inflation SLR is increased and it helps to decrease bank credit and to ensure solvency of commercial banks. Increase in SLR compels commercial banks to invest in government securities. While reserves under CRR is kept in the Central Bank, under SLR it is kept in the commercial bank itself. Further, while CRR is cash reserves, SLR can be in the form of cash, gold or securities. When CRR regulates the flow of money in the economy, SLR ensures the solvency of the banks.

iii) Open Market Operations -***Open market operations means the sale and purchase of government securities and bonds by the central bank***. When there is inflation the government securities are sold via commercial banks to the public such that a certain amount of bank deposits is transferred to the central bank as the public purchase government securities. As a result, the credit creation capacity of the commercial banks reduces.

Selective or Qualitative credit control measures

Under this method, extension of credit to essential purposes is encouraged and to non-essential purposes is discouraged. Hence these methods not only prevent the flow of credit into undesirable channels but also direct the flow of credit to useful channels. The important selective credit control measures are

i) Margin Requirements - Margin means that proportion of the value of security against which loan is not given. In other words 'Margin' refers to the difference between market value of securities and the amount of loan granted against these securities. For productive purposes margin requirements will be less.

ii)Regulation of Consumer Credit - Under this method the central bank lay down terms and conditions for the proper regulation of consumer credit given by the commercial banks of a country. Regulation of consumer credit restricts the amount of credit that might be given by commercial banks, restricts the time that might be available for repaying the loan, fixing the down payments etc.

iii) Moral suasion – These are the informal request by the central bank to commercial banks to contract credit in times of inflation and to expand credit in times of depression. The central bank issues periodical letters to commercial banks and discussions are held with authorities of commercial banks in this respect.

iv) Direct action – Central bank take direct action against erring banks. It may involve refusal by the central bank to rediscount bills or cancellation of license.

2. Fiscal Policy measures

These are the measures taken by the government to control the aggregate demand in the economy. The main instruments of fiscal policy are i) public revenue ii) Public expenditure iii) Public borrowing

i) Public revenue – The main source of public revenue is tax. When there is inflation the government want to reduce the total spending in the economy and hence tax is increased. Increase in direct taxes decreases the disposable income of the people and hence they spend less money.

ii) Public expenditure – During inflation the government cut down its expenditure on developmental activities and welfare programmes. This reduces government demand for goods and services as well as private income. When the government spend less money, income of the individuals decreases. Hence aggregate demand decreases.

iii) Public borrowing – When there is inflation the government will delay the repayment of public debt. At the same time the government should borrow more money from the public.

3. Other measures

Other measures include the measures taken by the government to increase the supply of goods and services, price control, wage control etc.

i) Increasing the supply of goods and services – When there is price rise government take various measures to increase the supply of goods and services. This can be done by importing essential products, banning the export of such items and by encouraging the production of essential commodities.

ii) Price control – Direct measures can be taken to control the price of goods and services. Essential commodities can be distributed through the public distribution system at reduced prices.

iii) Wage control – Wage control helps to prevent the escalation of cost of production during inflation and thus cost push inflation can be controlled.

Repo rate and Reverse repo rate

Repo rate is the rate at which RBI provides overnight liquidity to banks against the collateral of government and other approved securities. In other words, it is simply the rate at which RBI lends short term funds to commercial banks when they are facing a financial crunch. In this case, a repurchasing agreement is signed by both the parties stating that the securities will be repurchased by the commercial banks on a later date at a predetermined price.

Bank rate and repo rate are not the same. In general, repo rate focuses on providing funds to banks for a very short period where as bank rate focuses on long term fund requirements of the commercial banks. In the case of bank rate there is no repurchasing agreement signed and bank rate is usually higher than Repo Rate.

Reverse repo is the rate at which the RBI absorbs liquidity on an overnight basis from commercial banks. In other words, when a commercial bank has excess funds, they can deposit the same in central bank and earn interest in the form of reverse repo rate. The Reserve bank uses this tool when it feels that there is too much money floating in the banking system. An increase in the reverse repo rate means that the banks will get a higher rate of interest from RBI. As a result, banks prefer to deposit their money to central bank.

Repo rate and reverse repo rates are also used to control money supply in the economy. When there is excess money supply in the economy these two rates are increased.

As on 5 November 2021 the important monetary rates in India are as follows

Bank rate	-	4.25 %
CRR	-	4 %
SLR	-	18 %
Repo rate	-	4 %
Reverse repo rate	-	3.35 %

Chapter 8

Business Financing

12.1 Sources of Capital

Companies raise funds from domestic sources and foreign sources. The important domestic sources are

i) Internal Self-Finance:

An important source is the saving of the unit itself. It may be the household, the business or the government.

The households save as well as invest and lends its surplus to other units via, financial institutions like banks, capital market etc. The savings of the business includes depreciation and the retained earnings. When it is not adequate it borrows from financial institutions. Government also finances a part of their investment from internally generated funds. These arise from the excess of tax and other income over consumption spending plus transfers.

An advantage of investment through internally generated funds is that it combines the acts of saving and investment. It helps to reduce the cost of borrowing.

ii) Equity, Debentures and Bonds:

A large part of finance for fixed investments [building, machines, etc.] comes from different types of equity or shares. These shares bear risks of different degrees and cater to the needs of different investors. The latest trend is to issue shares in small denominations of ten rupees so as to enable the largest number of people to participate in providing long-term finance.

Companies also get long-term finance through the issues of debentures and bonds. These are debt (loans) instruments. The buyers of those debentures and bonds are the creditors of companies. They get a fixed rate of interest on the money invested in these securities.

iii) Public Deposits:

Another source is public deposits. It is also a debt-instrument, mostly for short-term finance. Under this system, people keep their money as deposit with these companies or managing authorities for a period of six months, a year, two years, three years or so. Depositors receive a fixed interest. This money is used by companies to meet their needs of working capital. However, this source of finance is unreliable because depositors can seek refund at any time.

iv) Loans from Banks:

Commercial banks also provide funds for meeting short-term needs or for working capital. Loans are given against the guarantee of government securities and stocks with companies. Loans are advanced in the form of overdraft and credit limit. Also commercial banks purchase debentures issued by the companies. They can earn fixed interest on such investment and at the time of need they can sell these debentures in the market and recover their money.

v) Indigenous Bankers: local bank

Indigenous bankers also advance financial help to a few large-scale industries, particularly during the time of stress, both for fixed capital and working capital. But mainly they have provided finance to small scale industries. These banks charge a very heavy rate of interest, thus making finance a costly affair.

vi) Development Finance Institutions: particular purpose - does provide

Development finance institutions cater to the needs of large and small industries. The new institutions supplying industrial finance are Industrial Development Bank of India, Industrial Finance Corporation of India, Industrial Reconstruction Bank of India, State Financial Corporations, and State Industrial Development Corporations. These institutions provide huge quantity of finances for setting up of new industries, for meeting their several needs and in several forms.

Shares and Bonds

When companies want to raise capital, they can issue Shares or bonds.

A share is a stake in the ownership of a company. It is a security that is also sometimes referred to as an equity. When a company issue shares, they are selling a certain amount of ownership in their company. An investor who buys the shares has a claim to the company's earnings and assets.

Some companies pay out a percentage of profits to investors in the form of dividends. However, companies are not obliged to pay dividends and they are not certain. Over time dividends can be increased, decreased or not declared at all.

When more shares are issued, future earnings must be shared among a larger pool of investors. More shares can cause a decrease in earnings per share (EPS), putting less money in owners' pockets. EPS is also one of the indicators that investors look at when evaluating a firm's health. A declining EPS number is generally viewed as an unfavourable development

Shares are perpetual investments and they do not have specific maturity period. The most attractive feature of stock issuance is that the money does not need to be repaid.

Bonds are a loan agreement that a company enters into with the investor. By buying a bond, an investor is lending money to a company for a pre-agreed period of time. For its part, the company agrees to pay back the money lent by the investor on a fixed date and to make regular interest payments during the period of the loan or in bulk at the time of maturity. When the bond reaches its maturity date, the company repays the investor.
Unlike shares, bonds are temporary investments which have fixed lifecycles. Although elements of the lifecycle may vary from bond to bond, the stages are the same from issue to maturity. Bond financing is often less expensive than equity.

Bond issuance enables corporations to attract a large number of lenders in an efficient manner. Record keeping is simple because all bondholders get the same deal. For any given bond, they all have the same interest rate and maturity date.

The following are the major differences between shares and bonds

Bonds	Shares
The investor lends money to the company	The investor owns part of the company
The Issuers of bonds are Govt. institutions, financial institutions, companies, etc.	Shares are issued by corporate enterprises
Risk is relatively low interest	Risk is very high some time get profit dividend

Bond holders get Interest, as a fixed payment	Shareholders get dividend, which is not guaranteed
Return is certain	Return is uncertain
As bondholders have a higher claim on assets, investors may still recover some of their initial capital	When a company is declared bankrupt Stocks will become worthless and investors may lose 100% of their capital
The capital is paid back in full to the investor at maturity	The amount of capital the investor gets back depends on the share price when the stocks are sold.
Maturity period is fixed	No maturity period for shares life long It is perpetual

Financial market

Money market and Capital market

A financial market deals with financial assets such as stocks, bonds, treasury bills, currencies etc. The two important components of a financial market are money market and capital market.

Money market

Money market deals with short term financial assets, that is, assets up to a maturity period of one year. The important instruments used in the money markets are deposits, collateral loans, bills of exchange, treasury bills, certificate of deposits etc. Institutions operating in money markets are central banks, commercial banks and acceptance houses etc.

Money markets provide a variety of services for individuals, corporates and government entities. Liquidity is often the main purpose for accessing money markets because short term debts are issued for the purpose of covering operating expenses or working capital for a company or government. . The money market plays a crucial role in ensuring companies and governments to maintain the appropriate level of liquidity.

Money Market is an unsystematic market, and so the trading is done off the exchange, i.e. Over The Counter between two parties by using phones, email, fax, online, etc. It plays a major role in the circulation of short-term funds in the economy. It helps the industries to fulfil their working capital requirement.

Functions of money market

The following are the important functions performed by the money market.

1. **Financing trade** -Money market finance internal and international trade. Finance is made available to the traders through bills of exchange, which are discounted by the bill market. The acceptance houses and discount markets help the traders in this process.

2. **Financing Industry -** Money market contributes to the growth of industries. Money market helps the industries in securing short-term loans to meet their working capital requirements through the system of finance bills, commercial papers, etc.

provide opportunity to investment

3. **Profitable Investment -** Money market enables the commercial banks and other financial institutions to invest their excess reserves in profitable way. Money market gives the opportunity to invest surplus funds in short term assets which are highly liquid.

saving is transferd to one sector to other

4.**Financial Mobility** - By facilitating the transfer for funds from one sector to another, the money market helps in financial mobility. Mobility in the flow of funds is essential for the development of commerce and industry in an economy.

5. **Equilibrium between Demand and Supply of Funds**-The money market brings equilibrium between the demand and supply of loanable funds.

6. **Economic growth**- Since money market helps in the development of trade, industry and agriculture, it promote overall economic growth.

more than one year

Capital Market

A capital market deals with long term financial assets. In other words a ***capital market is a financial market in which long-term financial assets are bought and sold***. Capital markets channel the wealth of savers to those who can put it to long-term productive use, such as companies or governments making long-term investment. The instruments which are traded in a capital market includes stocks, bonds, debentures etc. whose maturity period is not limited up to one year or sometimes the securities are irredeemable (no maturity). The Capital Market works under full control of Securities and Exchange Board to protect the interest of the investors. Capital markets are risky markets and are not usually used to invest short-term funds.

Newly issued shares

A capital market is broadly divided into two major categories: Primary Market and Secondary Market. A market where fresh securities are offered to the public for subscription is known as Primary Market where as a market where already issued securities are traded among investors is known as Secondary Market. share

Functions of capital market

1. Allocative function- The capital market functions as a link between savers and investors. It plays an important role in mobilising the savings and diverting them in productive investment. In this way, capital market plays a vital role in transferring the financial resources from surplus and wasteful areas to deficit and productive areas, thus increasing the productivity and prosperity of the country.

2. Encourages Saving- With the development of capital, market, the banking and non-banking institutions provide facilities, which encourage people to save more.

3. Encourages Investment - The capital market facilitates lending to the businessmen and the government and thus encourages investment. It provides facilities through banks and nonbank financial institutions. Various financial assets, *e.g.*, shares, securities, bonds, etc., induce savers to lend to the government or invest in industry. With the development of financial institutions, capital becomes more mobile, interest rate falls and investment increases.

4. Promotes Economic Growth-Various institutions of the capital market, like nonbank financial intermediaries, allocate the resources rationally in accordance with the development needs of the country. The proper allocation of resources results in the expansion of trade and industry in both public and private sectors, thus promoting balanced economic growth in the country.

5 Indicative Function– A Capital Market acts as a barometer showing not only the progress of a company but also of the economy as a whole through share price movements.

6. Liquidity function – In a developed capital market, securities can be purchased and sold without any delay and hence it ensures liquidity in the economy.

Major differences between Money Market and Capital Market

The following are the important difference between money market and capital market

1. The place where short-term marketable securities are traded is known as Money Market where as in Capital Market, long-term securities are created and traded. Capital Market is well organised which Money Market lacks.

2. The instruments traded in money market carry low risk, hence, they are safer investments, but capital market instruments carry high risk.

3. The major institutions that work in money market are the central bank, commercial bank, non-financial institutions and acceptance houses. On the contrary, the major institutions which operate in the capital market are stock exchange, commercial bank, non-banking institutions etc.

4. Money market full fills short term credit requirements of the companies such as providing working capital to them. As against this, the capital market tends to full fill long term credit requirements of the companies, like providing fixed capital to purchase land, building or machinery.

5. Capital Market Instruments give higher returns as compared to money market instruments. More risk

Return low
Less risk

6. Maturity of Money Market instruments is one year or less, but Capital Market instruments have a life of more than a year as well as some of them are irredeemable in nature.

7. Money market is unsystematic in nature where as a capital market is systematic in nature.

8. While money market helps to increase the liquidity in the economy capital helps in the mobilisation of savings in the economy.

Stock market

Share buying selling

The stock market refers to the collection of markets and exchanges where regular activities of buying, selling, and issuance of shares of publicly held companies take place. According to the Securities Contracts (Regulation) Act, 1956, "Stock Exchange means an association, organization or body of individual whether incorporated or not, constituted for the purpose of assisting, regulating, or controlling the business of buying, selling or dealing in securities. In short, stock market is an institution which provides a platform for buying and selling of existing securities. While both terms - stock market and stock exchange - are used interchangeably, the latter term is generally a subset of the former. If one says that he trades in the stock market, it means that he buys and sells shares/equities on one (or more) of the stock exchange(s) that are part of the overall stock market.

While today it is possible to purchase almost everything online, there is usually a designated market for every commodity. A stock market is a similar designated market for trading various kinds of securities in a controlled, secure and managed environment. Since the stock market brings together hundreds of thousands of market participants who wish to buy and sell shares, it ensures fair pricing practices and transparency in transactions.

As a primary market, the stock market allows companies to issue and sell their shares to the common public for the first time through the process of initial public offerings (IPO). This activity helps companies raise necessary capital from investors. A listed company may also offer new, additional shares through other offerings at a later stage. Following the first-time share issuance called the listing process, the stock exchange also serves as the trading platform that facilitates regular buying and selling of the listed shares. This constitutes the secondary market. As almost all major stock markets across the globe now operate electronically, the exchange maintains trading systems that efficiently manage the buy and sell orders from various market participants.

The following are some of the important functions of stock market.

- **Providing Liquidity and Marketability to Existing Securities**: The basic function of a stock market is the creation of a continuous market where securities are bought and sold. It gives investors the chance to disinvest and reinvest. This provides both liquidity and easy marketability to already existing securities in the market.
- **Pricing of Securities**: Share prices in stock market are determined by the forces of demand and supply. A stock exchange is a mechanism of constant valuation

through which the prices of securities are determined. Such a valuation provides important instant information to both buyers and sellers in the market.

- **Safety of Transaction**: The membership of a stock market is well- regulated and its dealings are well defined according to the existing legal framework. This ensures that the investing public gets a safe and fair deal on the market.
- **Contributes to Economic Growth**: A stock market is a market in which existing securities are resold or traded. Through this process of disinvestment and reinvestment savings get channelized into their most productive investment avenues. This leads to capital formation and economic growth.
- **Spreading of Equity Cult**: The stock market can play a vital role in ensuring wider share ownership by regulating new issues, better trading practices and taking effective steps in educating the public about investments.
- **Providing Scope for Speculation**: The stock market provides sufficient scope within the provisions of law for speculative activity in a restricted and controlled manner. It is generally accepted that a certain degree of healthy speculation is necessary to ensure liquidity and price continuity in the stock market

NSE

The National Stock Exchange of India Limited (NSE) is the leading stock exchange of India, located in Mumbai. NSE was established in 1992 as the first dematerialized electronic exchange in the country. It is the world's 10th-largest stock exchange according to May 2021 data. It was recognised as a stock exchange by SEBI in April 1993 and commenced operations in **1994** with the launch of the wholesale debt market, followed shortly after by the launch of the capital market segment.

In 1996, the NSE was the first exchange in India that planned to trade derivatives specifically on an equity index. In February 2000, the NSE started an Internet trading system. NSE provides a trading platform for various types of securities for investors under one roof - equity, debentures, central and state government securities, Treasury bills, commercial papers, certificate of deposits, mutual fund units etc.

BSE

oldest stock exchange

BSE (formerly known as Bombay Stock Exchange) was started in 1875. However, in 1850s, five stock brokers gathered together under a Banyan tree in front of Mumbai Town Hall. A decade later, the brokers moved their location to under another banyan trees at the junction of Meadows Street. With a rapid increase in the number of brokers, they had to shift places repeatedly. At last, in 1874, the brokers found a permanent location, the one that they could call their own. The brokers group became an official organization known as "The Native Share & Stock Brokers Association" in 1875. BSE is Asia's first & the

Fastest Stock Exchange in world with the speed of 6 micro seconds and one of India's leading exchange groups. Over the past 145 years, BSE has facilitated the growth of the Indian corporate sector by providing it an efficient capital-raising platform

STOCK EXCHANGE INDICES

Stock index

Stock market indices are the barometers of the stock market. They mirror the stock market behaviour. With some 7,000 companies listed on the Bombay stock exchange; it is not possible to look at the prices of every stock to find out whether the market movement is upward or downward. The indices give a broad outline of the market movement and represent the market. Some of the stock market indices are BSE Sensex, BSE200, Dollex, NSE-50, CRISIL-500, Business Line 250 and RBI Indices of Ordinary Shares. Mr. Charles Dow created the first stock market index known as the Dow Jones index back in May of 1896.

NIFTY

50 comps share price

NIFTY is a market index introduced by the National Stock Exchange. It is a blended word – National Stock Exchange and Fifty coined by NSE on 21st April 1996. NIFTY 50 is a benchmark based index and also the flagship of NSE, which showcases the top 50 equity stocks traded in the stock exchange out of a total of 1600 stocks.

These stocks span across 12 sectors of the Indian economy which include – information technology, financial services, consumer goods, entertainment and media, financial services, metals, pharmaceuticals, telecommunications, cement and its products, automobiles, pesticides and fertilizers, energy, and other services.

NIFTY contains a host of indices – NIFTY 50, NIFTY IT, NIFTY Bank, and NIFTY Next 50. Nifty is owned by India Index Services and Products Ltd. (IISL). It is calculated using the free float market capitalization weighted method where the level of index reflects the total market value of the stocks relative to a particular base period. The base period selected for calculating Nifty50 index is the close price on Nov 3, 1995. The base value has been set at 1000.

SENSEX

The BSE SENSEX (also known as the S&P Bombay Stock Exchange or Sensitive Index or simply the SENSEX) is a free-float market-weighted stock market index of 30 well-established and financially sound companies listed on Bombay Stock Exchange. These 30 companies are known as Blue chip companies. The 30 component companies which are

some of the largest and most actively traded stocks are representative of various industrial sectors of the Indian economy. Published since 1 January 1986, the S&P BSE SENSEX is regarded as the pulse of the domestic stock markets in India. The base value of the SENSEX was taken as 100 on 1 April 1979 and its base year as 1978– 79. On 25 July 2001 BSE launched DOLLEX-30, a dollar-linked version of the SENSEX.

Historically Sensex used the weighted market capitalization methodology, but from September 1, 2003, it shifted to Free Float Market Capitalization methodology. All the major indices in the world use the same methodology. The performance of the 30 selected key stocks directly reflects the level of the index.

If a person wants to trade in the stock market, he must obtain a demat and trading account.

Demat Account

Share buy - acct

Demat account is used to hold the shares purchased in digital or electronic form. During online trading, shares are bought and held in a Demat account, thus facilitating easy trade for the users. A Demat account holds all the investments an individual makes in shares, government securities, exchange-traded funds, bonds and mutual funds in one place. At any point or time, Demat account will show the shares and securities that a person is currently holding. In other words, it is a storage space to hold the shares and securities purchased. It is only a repository. It is similar to a bank account in which we hold deposits with the bank and the record of debit/credit balances are maintained in a bank passbook. In the same way, when we purchase or sell shares, it will be credited or debited to/from our Demat Account respectively.

Dematerialisation is the process of converting the physical share certificates into electronic form, which is a lot easier to maintain and is accessible from anywhere throughout the world. An investor who wants to trade online needs to open a Demat with a Depository Participant (DP - broker). The purpose of dematerialisation is to eliminate the need for the investor to hold physical share certificates and facilitating a smooth tracking and monitoring of holdings.

Demat account is an easy and convenient way to hold securities. It is safer than paper-shares and reduces paperwork for transfer of securities. It also reduces transaction cost and a single account can hold investments in both equity and debt instruments. Another advantage is that a person can trade from anywhere.

Trading Account

A trading account is used to buy and sell shares and securities in the stock market. A trading account provide an interface to buy and sell shares from the stock market. Previously, the stock exchange functioned on the open outcry system. In this, the traders used hand signals and verbal communication to convey their buying/selling decisions. Soon after the stock markets adopted the electronic system, trading accounts replaced the open outcry system. Most commonly, trading account refers to a day trader's primary account. These investors tend to buy and sell assets frequently, often within the same trading session.

In the online method, the buyers and sellers don't have to be physically present at the stock exchange to place orders. Instead, they open a trading account with a registered stock market broker; who conducts trading on their behalf. Each trading account has a unique trading ID which is utilised to perform online transactions. Nowadays, brokers provide facilities to the investors to perform transactions by themselves.

A trading account acts like a link between demat account and bank account of an investor. When an investor wants to buy shares, he places an order through his trading account. The said transaction goes for processing in the stock exchange. Upon execution, the required number of shares get credited into his demat account and a proportionate sum gets deducted from his bank account.

A similar kind of process is followed in order to sell equity shares. The investor places a sale order with the help of his trading account. It goes for processing in the relevant stock exchange. When the order is executed, the required number of shares are debited from his demat account and a proportionate sum gets credited to his bank account. At any point of time, a trading account will show the transactions we carried out in the stock market.

Thus, if we want to trade in the stock market, we need both the accounts. To open these accounts, documents like proof of identity, address proof, Pan card etc. are needed.

> **'Shares'** usually refers to units of ownership in a specific company – for example, you could say that you own ten Amazon shares.
>
> **Stocks'** is generally used to refer to portions of ownership of multiple companies – for example, you could say that you own stock in Amazon and Microsoft. That is, total amount of shares owns in more than one company.
>
> **'Equity'** is the term used for total ownership stake in the company – for example, if a company had 10,000 shares, and you owned 1000 of them, you could say that you held a 10% equity stake in that company.
>
> Usually, these terms are used interchangeably.

Chapter 9

International Trade

International trade or foreign trade means trade between countries. In other words, it is the exchange of goods or services between two or more countries. The branch of economics which deal with foreign trade is International Economics.

International Economics deal with the economic and financial transactions among nations. It analyses the flow of goods, services, payments and money between a nation and the rest of the world. The policies which regulate these policies and their effect on economic welfare of the nation are also coming under International Economics. More specifically, it deals with international trade theory, international trade policy, the balance of payments and foreign exchange markets.

Advantages and disadvantages of foreign trade

The following are the important advantages of foreign trade.

1. Optimal use of natural resources: International trade helps each country to make optimum use of its natural resources. Each country can concentrate on production of those goods which are advantageous to them. Therefore, wastage of resources is avoided.

2. Availability of all types of goods: It enables a country to obtain goods produced all over the world. The country may not be producing these good because of their technological problems or because of higher cost when it is produced in the domestic country.

3. Specialisation: Foreign trade leads to specialisation as the country produce only those goods, where the production of such goods has certain advantages. The country can enjoy the benefits of division of labour

4. Advantages of large-scale production: Due to international trade, goods can be produced for home consumption as well as for export. This enables country to produce in large scale and enjoy the benefits of largescale production.

5. Stability in prices: International trade helps to avoid wild fluctuations in prices by making the goods freely available all over the world.

6: Establishment of new industries: The importing of machinery and technology enable the country to start new industries.

7. Increase in efficiency: Due to international competition, the producers in a country attempt to produce better quality goods and to minimize the cost. This increases efficiency and productivity.

8. Development of the means of transport and communication:

International trade requires the best means of transport and communication. Because of this, countries develop better transport and communication facilities.

9. International co-operation and understanding: The people of different countries come in contact with each other. Commercial integration amongst nations of the world encourages exchange of ideas and culture. It creates cooperation, understanding, cordial relations among various nations.

10. **Discouragement to Monopolies:** International trade discourages the formation of monopolies in a country. If certain business units raise the prices through monopoly practices, the government imports those goods to reduce the prices in the country.

11. **Better Employment Opportunities:** As the Foreign trade expands, it creates jobs and provides better employment opportunities for the people both in and outside the country.

Disadvantages

1. A threat to domestic industries: International trade has an adverse effect on the development of home industries. It poses a threat to the survival of infant industries at home. Due to foreign competition and unrestricted imports, the upcoming industries in the country may suffer.

2. Economic dependence: Underdeveloped nations have to rely on developed countries for their economic growth. Such dependency often results in economic exploitation.

3. **Misuse of natural resources :** Constant and excessive exports can exhaust the natural resources in a country. If not controlled the country may suffer in the long run.

4. **Endangers independence:** Foreign trade encourages slavery. It impairs the economic independence of the poor nations.

5. **Import of harmful goods: Through international trade harmful goods may be imported and it may** adversely affects health and well being of the people.

6. **Evil Effects of Dumping:** Sometimes, certain countries use international trade to dump their goods on other countries with a view to cheapen the value of the other country's goods.

7 Against national Defence: It is argued that a nation which depends on foreign sources of supply lacks defence during the war. During war, they may not be able to import goods.

Theories Of International Trade

Absolute Advantage Theory

According to Adam Smith the basis of international trade is absolute cost advantage. Suppose there are two commodities and two countries which produce these commodities. One country is efficient in the production of one commodity and thus it has an absolute advantage in the production of this commodity. The other country has an absolute advantage over the production of the other commodity. Then the countries will specialise and produce that commodity upon which they have an absolute advantage. They will export this commodity to the other country. By this process resources are utilised in the most efficient way and output of both the countries will increase. From this mutual trade both the countries will benefit.

The absolute advantage theory can be explained with the help of an example.

	USA	UK
Number of Units of wheat per unit of labour	10	5
Number units of cloth per unit of Labour	3	8

In the above example USA has an absolute advantage over the production of wheat over UK. Because it is able produce 10 units of wheat with one unit of labour. But UK can produce only 5 units. Similarly, UK has an absolute advantage over the production of cloth. Hence, US will produce and export wheat to UK and UK will produce and export cloth to US. Both the countries will gain from international trade.

This kind of production leads to specialisation and division of labour. But according to Adam Smith division of labour is limited to the size of the market. When there is international trade, there is ample scope for division of labour because size of the market increases substantially. However, Adam Smith theory of absolute advantage explain only one aspect of trade. It is too narrow in its scope.

Comparative Advantage Theory

Comparative cost advantage theory was developed by David Ricardo in 1857. Later it was refined by J S Mill, Marshall and others. According to Ricardo, even in the case of a country for which there is no absolute advantage for both the commodities, it can gain from international trade. In this situation, the country should specialise in the production and export of the commodity in which its absolute disadvantage is smaller and import the commodity in which its absolute disadvantage is greater. In other words, a country should specialise in the production of that commodity in which it is more efficient and leave the production of the other commodity to the other country.

The Ricardian theory is based on the following assumptions.

1. There are only two countries and two commodities
2. There are no barriers in international trade
3. There is no transport cost
4. Labour is the only component of cost of production
5. There is perfect competition and full employment
6. Labour is homogeneous
7. Labour is perfectly mobile within the country
7. Goods are exchanged according to the relative amount of labour embodied in them.

Ricardo in his two commodity, two country model taken cloth and wine as commodities and England and Portugal as two countries.

Country	No. of units of labour Per unit of cloth	No. of units of labour per unit of wine	Exchange ratio
England	100	120	1 wine = 1.2 cloth
Portugal	90	80	1 wine = 0.88 cloth

The above example shows that Portugal has an absolute advantage in the production of both the commodities. However, a comparison of the ratio of the cost of wine production with ratio of the cost of cloth production in these two countries reveals that Portugal has a higher advantage in the production of wine. Hence it will specialise in wine production and produce wine. At the same time, England has a comparative advantage in cloth production and it will produce cloth. England can import wine from Portugal and Portugal can import cloth from England. Both the countries have mutual gain from trade as explained below.

In the absence of trade, one unit of wine commands 1.2 units cloth in England and 0.88 units of cloth in Portugal. When trade takes place, Portugal will gain if it can get anything

more than 0.88 units of cloth for one unit of wine. Similarly, England will gain if it has to sacrifice anything less than 1.2 unts of cloth. Therefore, any exchange ratio between 0.88 and 1.2 unts of cloth for one unit of wine will bring a gain for both the countries.

Criticism

Most of the assumptions of the theory are its limitations. The following are the important criticisms against comparative cost theory.

1. Labour is not the only element of cost.
2. Exchange ratio is not always fixed according to the cost ratios. Demand and supply play an important role in fixing the price.
3. The assumption of full employment and perfect competition are not valid.
4. The assumption of free trade (trade without barriers) is highly unrealistic.
5. According to Graham if one country is very small and other country is big complete specialisation may not be possible. The big country cannot sell its entire surplus to the small country.
6. The theory of comparative cost gives the limit within which exchange ratio will be fixed. It does not say how the exact point within these limits is determined.

The Heckscher-Ohlin Theorem or Factor Endowment Theory

The factor endowment theory was originally developed by Eli Heckscher in 1919. Later in 1935 it was refined by his student Bertil Ohlin. Hence the theory is popularly known as Heckscher-Ohlin Theorem. It is a two-country two-commodity model. The following are the important assumptions of the model.

1. There is perfect competition in the factor market and product market.
2. Factors of production are perfectly mobile within the country but immobile between the countries.
3. Factors production are homogeneous.
4. There is full employment.
5. There is free trade between countries.
6. There is no transport cost.
7. Technology remains the same in both countries.

The classical theory showed that the basis of international trade was comparative cost differences. But it did not explain the reason for comparative cost differences. The Heckscher-Ohlin theorem tried to explain the causes of comparative cost differences that exist internationally. According to the theorem, the differences in comparative advantage among nations is mainly due to the differences in relative factor abondance or factor endowments.

Heckscher-Ohlin theorem can be stated as follows. A country will produce and export that commodity whose production requires the intensive use of the nation's relatively abundant and cheap factor and import the commodity whose production requires the intense use of relatively scarce and expensive factor. In other words, relatively labour abundant country will export the relatively labour-intensive commodity and import the relatively capital-intensive commodity.

In the Heckscher-Ohlin theorem factors of production are considered as abundant or scarce in relative terms and not in absolute terms. For example, a country will be considered as capital abundant only if the ratio of capital to other factors is higher when compared to other countries.

Country A	Supply of labour	= 50
	Supply of capital	= 40
	Capital- labour ratio	= 0.8
Country B	Supply of labour	= 16
	Supply of capital	= 20
	Capital-labour ratio	= 1.25

In the above example, country A has more capital in absolute terms but country B is endowed with or abundant in capital because the ratio of capital to labour is high in country B.

Factor Price Equalisation Theorem

Factor price equalisation theorem is a corollary of Heckscher-Ohlin theorem. It was proved by Paul Samuelson and hence it is called Heckscher-Ohlin-Samuelson theorem.

The theorem state that free international trade will equalise factor prices between countries relatively and absolutely and this serve as a substitute for international factor mobility. In a country, international trade increases the demand for abundant factors because specialisation takes place on the basis of factor endowments or abundance. Therefore, the

prices of the abundant factors increase. Similarly, the demand for the scarce factors decreases and hence their prices also decrease. Thus, when a country export goods containing a large proportion of the relatively abundant and cheap factors and import goods containing a large proportion of scarce factors, it may act as substitute for inter-regional factor movements and lead to factor price equalisation.

Merits of Heckscher-Ohlin theory

1. Heckscher-Ohlin theory provides a more comprehensive and satisfactory explanation for foreign trade.

2. Heckscher-Ohlin theory explain the reason for comparative cost differences between nations in terms of factor endowments.

3. Heckscher-Ohlin theory is formulated within the framework of the general equilibrium.

4. The Heckscher-Ohlin theory highlights the role of relative prices of factors in determining the trade flow.

5. Heckscher-Ohlin theory highlights the impact of trade on product and factor prices.

Effects of International Trade

According to Heckscher-Ohlin theorem, international trade has the following effects.

1. Equalisation of factor prices: Since specialisation takes place according to factor endowments, it equates factor prices between countries.

2. Equalisation of Commodity prices: International trade leads to the movement of goods from those areas where they are abundant to areas where they are scarce. This would equalise the commodity prices.

Balance of Payments

Balance of payments is a systematic record of all economic transactions of a nation with the rest of the world for a specific period of time. Usually, time period is taken as one year. The main purpose of balance of payments is to inform the governments regarding the international currency position of the nation and to help in the formulation of policies accordingly. Balance of payments is also useful to banks, firms and individuals who directly or indirectly involved in international trade and finance.

It is obvious that during a period of time millions of transactions take place between one nation and with the rest of the world. Therefore, all these transactions cannot appear individually in the balance of payment statement. As a summary statement, balance of payments aggregates all these transactions under different heads.

Balance of Trade and Balance of Payments

It is meaningful to distinguish between balance of payments and balance of trade. Balance of trade includes only those transactions which are involved in the exporting and importing of visible items (goods). It does not include invisible items such as various kinds of services (shipping, banking, insurance), payment of interest and dividend etc. On the other hand, balance of payments includes both visible and invisible item. Therefore, balance of payments gives a better picture of a country's external balance.

Components of Balance of Payments

Balance of payments accounting follows double-entry book-keeping system. That means each transaction will result in a credit entry and debit entry of equal size. Therefore, balance of payments will always balance. That is, total amount of debit will be equal to total amount of credit. Sometimes, an item called errors and omissions will be added to balance the balance of payments.

Usually, international transactions are classified under the following heads:

1. Current Account 2. Capital Account 3. Unilateral payments Account
4. Official Reserve Assets Account

Current Account

Current account consists of two major items i) merchandise (visible) exports and imports ii) invisible exports and imports.

Merchandise exports and imports: Merchandise exports, that is sale of goods abroad are credit items and merchandise imports, that is purchase of goods from abroad are debit items. Merchandise exports and imports are the most important international transactions of most of the countries.

Invisible exports and imports: Invisible exports are credit entries and imports are debit entries. Invisible exports mean sale of services like transport, insurance, foreign tourist expenditure in the home country and interest received on loan and dividend on investment abroad etc.

Invisible imports include purchase of services like transport and insurance, tourist expenditure abroad and payment on foreign loans and foreign investments etc.

Capital Account

The capital account includes short term and long-term capital transactions. Capital inflows are coming as credit entries and capital outflows as debit entries. For example, when a Japanese firm invest 100 million in India, there will be an entry under credit for India and an entry under debit for Japan. However, the payment for capital services like interest on loan and dividend payments for investment are included in current account. The following are the three major items coming under capital account.

1. Loans and borrowings – It includes all types of loans from both the private and public sectors located in foreign countries.
2. Investments – These are funds invested in the corporate stocks by non-residents.

The flow of funds from and to foreign countries through various investments in real estates, business ventures, foreign direct investments etc is monitored through the financial account. This account measures the changes in the foreign ownership of domestic assets and domestic ownership of foreign assets

Unilateral Transfers Account

Unilateral Transfer means the one-way transfer of an item from one person to another. Such one-way transfers are without any expectations of anything in return. This is an important item of balance of payments. In a broader sense it is a part of current account. The following are the important items coming under unilateral transfers.

- Payments or remittances from immigrants to their home country.
- Humanitarian aid.
- Aid by one country to another. Usually, the aid is from developed or prosperous nations to less developed nations.
- Contribution to charitable institutions.
- Membership payment to international agencies.
- Gift from one county to another. This gift could be from a person, business or government.

The official reserve account

The official reserve account is a subdivision of the capital account. It is the foreign currency and securities held by the government, usually by its central bank, and is used to balance the payments from year to year. The official reserves increases when there is a trade surplus and decreases when there is a deficit. Sometimes the central bank will use it to intervene in the foreign exchange market to set the exchange rate to some desired level.

Balance of Payments Disequilibrium – Deficit

The balance of payments is in disequilibrium when it shows a surplus or deficit. When the demand for foreign exchange exceeds the supply of foreign exchange, there is deficit in balance of payments. There are a number factors responsible for a disequilibrium or a deficit in balance of payments.

Economic Factors

The following are the important economic factors which lead to balance of payment disequilibrium.

i) Development disequilibrium: Largescale development expenditure may increase the purchasing power of the people and they demand more imported items. Besides, developing countries may import capital goods like machinery and equipment for their economic development. This also increase their import bill and result in a deficit.

ii) Cyclical Disequilibrium: Cyclical fluctuation create and boom and depression. When there is boom import may increase more than export and it create a deficit in balance of payments.

iii) Secular Disequilibrium: In developed countries disposable income and aggregate demand will be very high. Wages may increase and the production cost also increases. This may result in high prices. Higher aggregate demand and higher prices lead to larger imports and deficit in the balance of payments.

Political Factors

Political instability in a country may adversely affect the capital flows and investments. It may lead to large capital outflows and less investment in the domestic country. This may create a deficit in balance of payments.

Social factors

Changes in tastes, fashion etc. may change consumption habits of the people and this may affect export and import as well as balance of payments.

Correction of Disequilibrium or Deficit

When there is deficit in the balance of payments, it has to be corrected. The following are the important measures to correct a balance of payments disequilibrium.

1. Automatic Correction: When there is a disequilibrium in the balance of payments, the economy will try to correct it automatically with the help of demand supply mechanism. For example, when there is a deficit, or the demand for foreign exchange exceeds it supply, the exchange rate will be get adjusted accordingly and there will be a fall in the external value of the domestic currency. This will increase the exports and correct the deficit.

2. Deliberate Measures: The three deliberate measures are a) Monetary measures b) trade measures and c) Miscellaneous measures

Monetary measures The important monetary measures are

i) Monetary contraction or expansion: Expansion or contraction of money supply can affect the balance of payments position. When there is a deficit, a contraction of money supply will decrease the purchasing power of the people and hence the aggregate demand as well as the price level falls. This reduces the imports. When there is a fall in the price in the domestic economy, it encourages export. Thus, deficit will be corrected.

ii) Devaluation: It means a deliberate reduction in the value of a currency by reducing the official rate at which it is exchanged for another currency. When there is a deficit in the balance of payments, devaluation of the currency encourages export and discourages import. Devaluation makes the domestic goods cheaper for the foreigners and foreign goods expensive for the people in the home country.

iii) Exchange control: Under exchange control, the government or the central bank will have the complete control over the foreign exchange earning of the country. The government will keep the entire foreign exchange earnings and this helps the government to control imports.

Trade Measures Trade measures are export promotion and import control.

i) Export promotion: Export can be encouraged by abolishing export duty, giving export subsidy and by providing facilities for export-oriented production.

ii) Import control: Imports can be discouraged by increasing import duties, through import quotas import licensing etc.

Miscellaneous Measures Miscellaneous measures include encouragement of foreign investment, promotion of tourism etc.

Devaluation

Devaluation means a deliberate reduction of the value of the domestic currency in terms of foreign currencies. A country which faces a serious problem of deficit in the balance of payments may resort to devaluation. This will stimulate their export and discourage import.

The working of devaluation can be explained with the help of the devaluation of Indian rupee in 1966.

Before devaluation the exchange rate was $1 = Rs. 4.76. Devaluation of the Indian Rupee was 36.5 per cent and it was 57.56 against dollar. Then the exchange rate became $1 = Rs.7.5. After devaluation import became costly. Before devaluation, a foreign commodity which cost $1 Abroad cost Rs.4.76 in India. But, after devaluation the same commodity cost $1 abroad but Rs.7.5 in India. Thus, imported goods became costlier in India. Similarly, export became cheaper after devaluation. Before devaluation a commodity which cost Rs.4.76 in India and $1 abroad now cost only $0.64 for the foreigners. This made Indian goods cheaper in the foreign market and encouraged export.

Limitations of Devaluation

1. The success of devaluation depends on reactions of other countries. If they retaliate by devaluing their currencies, devaluation will not be successful.

2. If prices in the domestic country increases at the same rate or at a higher rate of devaluation, it will not will not increase export or decrease import.

3. The success of devaluation depends also depends on the elasticities of demand for export and import. According to the Marshall-Lerner condition devaluation will be successful only if the sum of elasticities of demand for exports and imports of the domestic country is greater than one.

4. In spite of an increase in the demand for a country's export due to devaluation, the extent of increase in exports depends on the exportable surplus or the quantity available for export.

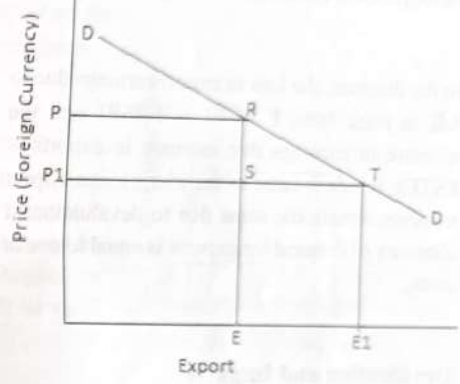
Devaluation and Elasticities of Demand for Exports and Imports

The success of devaluation depends on the elasticities of demand for exports and imports.

Devaluation and Exports

The effects of devaluation on exports depends on the elasticity of demand for a country's exports by the foreigners. The following three situations makes this clear.

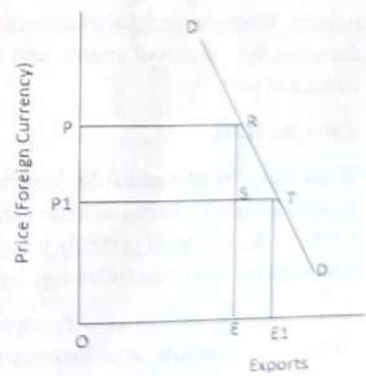
1. More elastic demand: When demand for exports is more elastic, a fall in price of exports in terms of foreign currency increases the export earnings and there will be a favourable effect on balance of payments. The following diagram illustrate this.



In the diagram price of exports is measured in terms in terms of foreign currency. When price of exports falls from P to P1 due to devaluation, exports increase from E to E1, The loss in export earnings due to fall in price is shown by the area PRSP1. But the increase in export earnings due to decrease in price as a result of devaluation is ESTE1, which is greater than the loss. Thus, there is favourable effect on balance of payments.

Less Elastic Demand

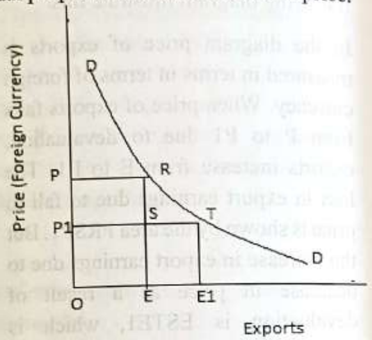
When price elasticity of demand is less than one, there will be a decrease in export earnings due to devaluation. Here, export will increase at a less proportion when compared to the decrease in price. In this case, when price falls from P to P1, the loss in earnings due to fall in price is PRSP1. But the increase in earnings due to the increase exports is ESTE1, this is less than the loss. Thus, devaluation will have an unfavourable effect on balance of payments when the elasticity of demand for export is less than unity.



Unit elastic Demand

When Demand for exports is unit elastic, devaluation will have no effect on the export earnings. Here the increase in exports will be in equal proportion to the decrease in price.

In the diagram, the loss in export earnings due to fall in price from P to P1 is PRSP1 and the increase in earnings due increase in exports is ESTE1, which is same as the loss. Hence, export earnings remain the same due to devaluation, if elasticity of demand for exports is equal to one or unity.



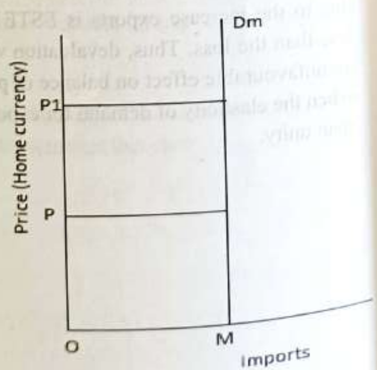
Devaluation and Imports

Devaluation increases the price of the imports in terms of home currency. As price in the international market remains the same, price in terms of foreign currency remains the same. The effect of devaluation on balance of payments depends on elasticity of demand for imports. When demand for imports is greater than zero, an increase in price of imports decreases the volume of imports and as result there will be a favourable effect on the balance of payments.

Zero elasticity

When elasticity of demand for imports is zero, imports will not change as the price of imports increases in terms of home currency. In other words, demand is perfectly inelastic. This situation is shown in the following diagram.

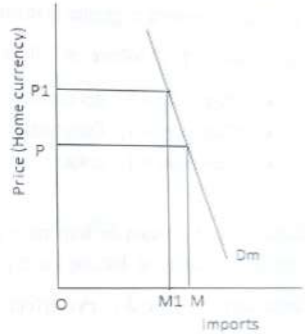
In the diagram, when price is P, imports is M. When price in terms of home currency increases to P1, that time also imports remains at M. Hence, there will not be any change in the import bill in terms of foreign currency. Therefore, there will not be any impact on balance of payments.



Less elastic demand

When the value of elasticity of demand is greater than zero but less than one, there will be a decline in imports due to an increase in the price of imports. But it will not be substantial. Hence, there will be some favourable effect on the balance of payments. This shown in the diagram.

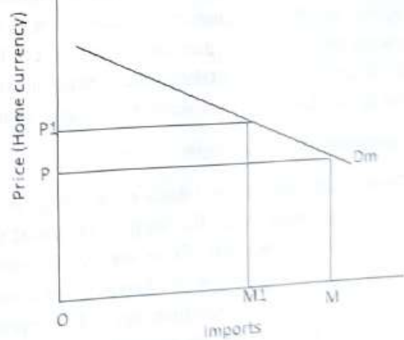
In the diagram, when price increases from P to P1, imports decrease from M to M1. When compared to the increase in price, the decrease in imports is proportionately very small. Thus, when demand for imports is relatively inelastic, the Impact of devaluation on balance of payments will be small but it is favourable.



More elastic demand

When demand for imports is more elastic, an increase in price causes a substantial decrease in the volume of imports and hence it has larger impact on import bill as well as balance of payments. The following diagram illustrate this.

In the diagram, when price increases from P to P1 imports decreases from M to M1. This is substantially a larger decrease in the volume of imports and hence the import bill also decreases substantially. Thus, to get a good result on balance of payments through devaluation, demand for imports should be more elastic.



The above analysis shows that the impact of devaluation on balance of paints depends on the combined effect of elasticity of demand for exports and imports. This combined effect is explained in Marshal-Lerner Condition.

Marshall-Lerner Condition

The Marshall-Lerner condition states that devaluation will improve the balance of payments of a country if the sum of elasticities of demand for a country' exports and its demand for imports is greater than one. In other words

$ex + em > 1$ where ex is the elasticity of export and em is the elasticity of import.

- If $ex + em > 1$, devaluation will improve current balance of payments position
- If $ex + em < 1$, Devaluation will deteriorate balance of payments
- If $ex + em = 1$, devaluation will have no effect on balance of payments

If elasticity of demand for imports is equal to zero, then also devaluation will be effective, provided elasticity of demand for exports is greater than one.

Devaluation – the J-curve effect and Currency Pass-Through

The J-curve shows the time path of trade flows after devaluation. It says that, devaluation will lead to an initial deterioration of the trade balance and this will be followed by a subsequent improvement. Empirical evidence proves this kind of behaviour of trade balance. That is, if the balance of trade is plotted against time, it will give a 'J' shaped curve showing the initial deterioration and the subsequent improvement.

One possible reason for the 'J' curve effect is currency invoicing. That is, if export contract is in terms of domestic currency and import contract in terms of foreign currency, this may happen. Another explanation is decision lags in forming new business connections and placing new orders. Further, another reason can be the time lag between the new orders placed and the payment takes place after devaluation. Production lag also can be a reason, that is the time taken to increase the output of goods for which demand has increased.

Currency Pass-Through

Theoretically it is assumed that a given change in exchange rate will bring a proportionate change in import prices. But, in practice import prices change less than proportionately, thus weakening the effect of devaluation. The extent to which the change in exchange rate leads to a change in export and import prices is called currency pass through relationship. This may be due to various time lags and other problems. Regarding the US it is estimated that for every 10 percent change in the value of dollar leads to only six percent change in import and export prices.

Free Trade Versus Protection

Free trade is a trade policy that does not restrict imports or exports. In other words, it refers to the trade that is free from all artificial barriers to trade like, tariffs, quota restrictions, exchange control etc. In the case of free trade the free market idea is applied to international trade. Protection on the other hand is the policy of protecting domestic industries against foreign competition by means of tariffs, subsidies, import quotas etc. It affects mainly the imports of a country. Government-levied tariffs are the chief protectionist measures. They raise the price of imported products, making them more expensive than domestic products.

Arguments for Free Trade

The following are the important arguments in favour of free trade

1. Better utilisation of resources: Free trade leads to the most economic utilisation of resources. The resources will be utilised in the production of those goods for which it is best suited.

2. Division of labour and specialisation: Under free trade Each country will specialize in the production of those goods in which it has a comparative advantage. This leads to large scale production, division of labour and efficiency.

3. Efficiency: There is intense competition under free trade and hence inefficient firms cannot survive. Therefore, firms try to increase their efficiency.

4. Dampen monopoly practices: Free trade jeopardise domestic monopolies by providing internationally available goods at lowest price.

5. Wide variety of goods: Free trade enables the consumers to avail all types internationally available goods at cheapest price.

6. Avoid corruption and red-tapism: Free trade is free from bureaucratic interferences and hence it avoids corruption and delay in taking trade related decisions.

7. Economic growth: Largescale production and division of labour leads to economic growth.

Arguments Against Free Trade

1. Threat to domestic industries: Because of free trade, imported goods become available at a cheaper price. Thus, an unfair and cut-throat competition develops between domestic and foreign industries. In the process, domestic industries are wiped out.

2. Harmful commodities: a country may have to change its consumption habits. Because of free trade, even harmful commodities (drugs, etc.,) enter the domestic market. To prevent this, restrictions on trade are required to be imposed.

3. The Unfair-Competition Argument: It is argued that free trade leads to competition among unequals. Developing countries cannot fairly compete with the developed countries. Their cost conditions may be different.

4. Job outsourcing leads to unemployment: Free trade allows businesses to move their production to a place where it is cheaper to produce. In countries where labour or production costs are high, the firms may outsource their work and this may lead to loss of employment domestic economy.

5. Degradation of environment: Emerging developing economies often may not have sufficient environment protection laws. Free trade leads to depletion of timber, minerals, and other natural resources because of their over exploitation.

6. Poor Working Conditions: Multi-national companies may outsource jobs to emerging developing countries without adequate labour protections. As a result, women and children often may involve in factory jobs in sub-standard conditions.

Arguments in Favour of Protection

When the domestic industries are threatened by foreign competition, nations may resort to protectionism to safeguard national interest. The following are the important arguments in favour of protection.

1. Infant industry argument: This argument says that when a new industry is launched, it must be protected from foreign competition. Already established companies will have certain advantages like economies of scale, experience, market power etc. If the infant is to compete with such a foreign competitor, it will be competition between unequals and that will lead to the destruction of the infant industry. That doesn't mean that it has to be protected for ever, but only during the blooming stage. According to protection policy "Nurse the baby, Protect the child and Free the adult". But some economists criticised that if protection is given to an infant, it will remain as an infant forever.

2. Strategic and Key industry argument: It is argued that a country should develop its own strategic and key industries. This is because the development of other industries and the development of the economy needs the output of these industries. Hence, we have to protect and develop such industries.

3. National Defence: If we depend on other nations for defence equipment it will be a foolishness because if they deny it at times when it is most urgent it will be a threat to the security of the nation. Hence defence industries should be protected and developed.

4. Diversification: A diversified industrial structure is necessary to maintain stability in the economy and to strengthen the economy.

5. Terms of trade argument: When a country protects its industries by imposing tariffs or quotas, it will restrict the imports and improve the terms of trade.

6. Improving balance of payments: When imports are restricted through protection policies, it will help to improve the balance of payments.

7. Anti-Dumping: Through dumping a foreign company may sell its product at very low prices in the home country and this may ruin the domestic industries. Once they get a monopoly over the product, they will increase the price and hence it will be harmful in the long run. By protective measures, a government can prevent dumping.

8. Employment argument: Restricting imports by way of protective measures encourages production in the home country and it will create more employment opportunities in the home country.

9. **Keeping money at Home:** When a commodity is imported, money from the home country is going abroad. On the other hand, when imports are restricted through protection, money can be kept in the home country itself.

10. Equalisation of costs of production: Imposition of import duties increases the price of foreign goods and thus it is argued that it will equalise the cost of the commodity in the home country and in the foreign market. But critics say that cost differences is the very basis of international trade.

11. Size of the home market: It is argued that protection will enlarge the market for agricultural products. This will happen because of the increase in the price of the farm products imported from foreign countries as well as the increase in the purchasing power of the workers who engage in the industrial sector by way of protection.

Arguments Against Protection

1. Protection is against the interest of the consumers as it increases the price of the imported products. Further, consumers are denied the opportunity for enjoying variety goods.

2. It discourages competition and hence compromises efficiency.

3. It encourages the growth of domestic monopolies because of the weakening of foreign competition.

4. Protection discourages innovations and cost reduction.

5. It leads to uneconomic utilisation of world's resources.

6. Protection may lead to trade wars and international conflicts among trading nations. When one country takes protective measures, others may retaliate.

Trade Barriers

Trade barriers refer to the government policies and measures which restrict the free flow of goods between the countries. Broadly, trade barriers are divided into two groups. They are tariff barriers and non-tariff barriers.

Tariff Barriers

Tariff barriers are duties or taxes imposed by the government of a country on its imports or exports. Tariff is an important measure of protection. But, recently, because of the involvement of WTO countries are reducing tariffs.

On the basis of origin and destination of goods, tariffs can be classified in to the following three categories.

i) Export duties: These are taxes levied on goods originated in the duty levying country (home country) and destined for some other countries.

ii) Import duties: These are taxes imposed on goods originated abroad and destined for the duty levying country (home country).

iii) Transit duty: These are taxes imposed on goods crossing the boarders of a country but these goods are originated from and destined for some other countries.

Based on the quantification of tariffs, these can be classified as

i) Specific duties: It is a fixed amount of duty imposed on each unit of the commodity exported or imported.

ii) Ad-valorem duties: These are duties levied as a fixed percentage of the value of the commodity imported or exported.

iii) Compound duties: When specific and ad-valorem duties are imposed on a commodity it is known as compound duties.

Based on the application of Tariffs between different countries, tariffs may be classified as

i) Single-column tariff: Under this type of tariff system, a uniform tariff is imposed on similar product irrespective of the country from which they are imported.

ii) Double-column tariff: In this type two rates are imposed on some commodities or all commodities.

iii) Triple-column tariff: In this system three rates of tariffs – general, the intermediate and preferential- are levied.

Based on the purpose they serve, tariffs can be classified as

i) Revenue tariff: When raising revenue is the only motive of imposing a tariff, it is called revenue tariff. Generally, this type of tariff rates will be low.

ii) Protective tariff: This type of tariff is imposed with an intention to protect domestic industries. Usually, the rates will be high.

iii) Countervailing and Anti-Dumping tariffs: Countervailing tariffs are imposed on those commodities which are heavily subsidised by the foreign governments. When foreign goods are sold in the domestic market at price lower than its cost of production, anti-dumping duties are imposed.

Effects of Tariff

The following are the effects of tariff on the economy.

i) Protective effect: When an import duty is imposed, imports become costlier. Hence, the demand for domestic goods will increase and it will protect the domestic industries.

ii) Revenue effect: A tariff will increase the revenue of the government if it does not completely stops the imports.

iii) Income and employment: A tariff will increase the demand for domestic goods. Hence it will increase production, employment and income in the home country.

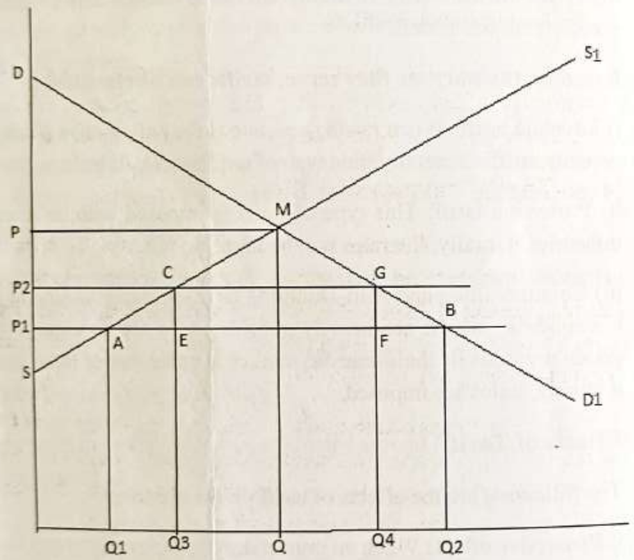
iv) Balance of payments effect: Tariff may help to improve the balance of payments as its restrict the imports.

v) Consumption effect: An import duty will increase the price of the commodities and hence it will reduce the buying capacity of the people.

vi) Competitive effect: An import duty protects the domestic industries and hence it may reduce competition in the economy. This can lead to inefficiencies.

vii) Redistribution effect: If the import duty increases the price of a domestically produced product, it leads to a redistribution of income in favour of the producers.

The effects of tariff in general can be explained with the help of the following diagram.



In the diagram, in the absence of foreign trade domestic demand curve DD1 and supply curve SS1 intersect at point M. The equilibrium price is P and the quantity demanded and supplied is Q. Suppose, foreign supply is perfectly elastic at price P1. Then under free trade, supply in the economy can be represented by the straight line P1B. Under free trade, the total consumption will be Q2 and out of this Q1 will be supplied by domestic producers and the quantity Q1Q2 will be imported.

Suppose the government imposes a tariff equal to P1P2. This increases the price from P1 to P2. At P2 price domestic demand falls to Q4. Because of the rise in price domestic supply increases from Q1 to Q3. The remaining part of the domestic demand Q3Q4 will be met by import.

Under free trade the total consumers surplus is DP1B. But under protection it decreases to DP2G. The total loss in consumer's surplus is P1P2GB and this is distributed in a number of ways. The tariff per unit is P1P2 and total import is Q3Q4. Therefore, the government will get a total revenue of CEFG from import duty.

Because of the imposition of the tariff, price increased from P1 to P2. Hence, the producers get additional benefits of P1P2CA. This is a transfer of income from consumers to producers and it is the redistributive effect of tariff.

Because of the tariff domestic supply increases from Q1 to Q3. ACE represents the sum of additional cost per unit of output This is the protective effect of the tariff. Similarly, consumption decreases from Q2 to Q4 and hence the loss in consumer's surplus is GFB. This is the consumption effect.

The actual loss to the economy because of the tariff is the sum of protective effect and consumption effect (ACE + GFB). The revenue effect (CEFG) and redistributive effects (P1P2C) are only a transfer of income from one group to another group.

Non-Tariff Barriers (NTBs)

Recently, non-tariff barriers are gaining popularity. Its impact is more on developing countries than developed countries. There are different types of NTBs. Hardcore NTBs include, Voluntary Export Restraints, Variable levies, Multi-Fibre Agreement restrictions and non-automatic licensing. Others include technical barriers, minimum pricing regulations and price surveillance.

1. Voluntary Export Restraints: A voluntary export restraint (VER) is a trade restriction on the quantity of a good that an exporting country is allowed to export to another country. This limit is self-imposed by the exporting country. They are highly discriminatory and the WTO is taking efforts to eliminate such restrictions.

2. Administered Protection: Administered protection encompasses a wide range of bureaucratic government actions. The important measures under administered protection include

a) Safeguards: A safeguard is a temporary import restriction that a country is allowed to impose on a product if imports of that product are increasing so as to cause, or threaten to cause, serious injury to a domestic industry that produces a similar or directly competitive product.

b) Health and product standards: The developed countries fix certain health and product standards which hinder the exports of developing countries because of added cost or technical requirements.

c) Customs Procedures: Customs procedures of many countries act as a trade barrier. For example, frequent changes in customs procedures of Japan acted as a barrier to export.

d) Licensing: Many countries use licensing as a measure to restrict trade, especially imports.

e) Monetary controls: Monetary controls are also employed to regulate imports. For example, RBI in 1990s took several measures which include a 25 percent interest rate surcharge on bank credit for imports.

f) Environmental protection laws: Many countries framed environment protection laws to restrict imports. For example, the US congress has passed legislation to prohibit the import of shrimp harvested with commercial fishing technology on the argument that it will threaten the survival of turtles.

g) Foreign Exchange Regulations: In some countries, the State monopolise foreign exchange and hesitate to release foreign exchange for imports.

Demerits of NTBs

NTBs are less transparent, difficult to identify and it is impossible to quantify its impacts. They are unfair because they do not treat exporters equally.

Quantitative Restrictions or Quotas

A quota represents a ceiling or limit on the volume of exports or imports. The following are the important types of import quotas.

1. The tariff or custom quota: Under this system, import of a commodity up to a specified quantity is allowed to be imported duty-free or at a special low rate of duty. But imports in excess of this fixed limit are charged a higher rate of duty. The tariff quota thus combines the features of a tariff with those of quota.

2. The Unilateral Quota: Under this system, a country places an absolute limit on the import of a commodity during a given period. It is imposed without prior negotiation with foreign governments.

3. The Bilateral Quota: Under this system, quotas are set through negotiation between the importing country and the exporting country.

4. The Mixing Quota: It is a type of regulation which requires producers to utilise a certain proportion of domestic raw materials along with imported parts to produce finished goods domestically. It thus sets limits on the proportion of foreign-made raw materials to be imported and used in domestic production.

5. Import Licensing: Under this, prospective importers are required to obtain a licence from the proper authorities for importing any quantity within the specified quotas. Licences are generally distributed among established importers keeping in view their share in the country's import trend.

Effects of quotas

The following are the important economic effects of quotas

1. Price effect: As quotas restrict imports and domestic supply, it increases prices in the domestic economy.

2. Consumption effect: Since quotas raises the price, it reduces consumption.

3. Balance of payments effect: As quotas restrict imports, balance of payments position will be improved.

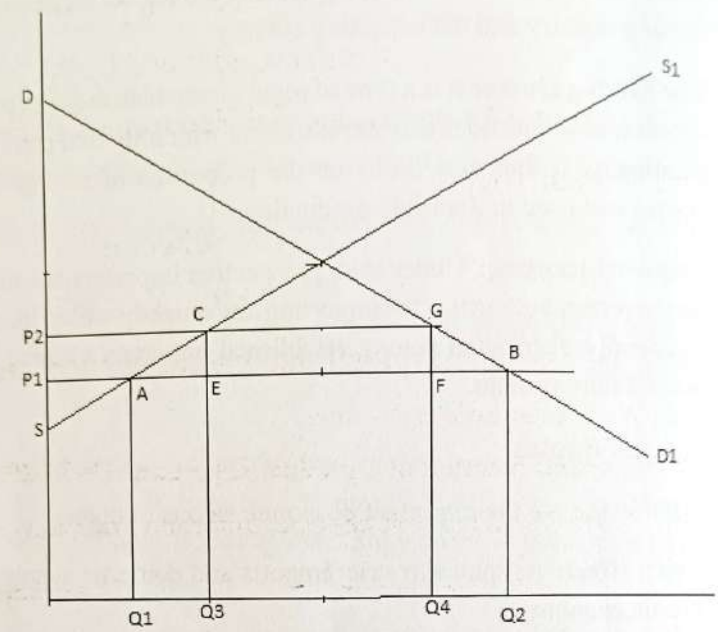
4. Protective effects: Quotas restrict imports and guard domestic industries from foreign competition.

5. Revenue effect: When quotas are administered by means of a license, government can get some revenue in the form of license fee.

6. Redistributive effect: If price increases because of quota restrictions, there will be some redistribution of income in favour producers.

Effects of quotas can be explained with the help of the following diagram similar to the case of tariff.

In the diagram DD1 is the domestic demand curve and SS1 is the domestic supply curve. The foreign supply is assumed to be perfectly elastic at price P1. At price P1 the domestic demand is Q2, out of which Q1 quantity will be supplied by domestic producers and Q1Q2 will be met by way of import.



Suppose the government fixes the import quota as Q3Q4. The reduction in supply increases the price to P2. At P2 price domestic suppliers are producing a larger quantity Q3.

Because of the increase in price from P1 to P2, producers get additional income equals to P1P2CA. This is the redistributive effect. ACE represents the protective effect (sum of additional cost per unit of output because of the increase in production) and BGF represents the consumption effect (loss in consumer's surplus due to quota restriction).

Tariff Versus Quotas

1. Quota is more effective in regulating trade. When a tariff discourages imports, quota directly restrict the quantity of imports.

2. Compared to tariff quotas are more precise and certain in action.

3. Quotas can be more easily imposed and more easily removed.

4. Quota can be used to prevent transmission of recession from foreign countries to home country. When there is recession prices in the foreign country decrease drastically and this nullify the effect of a tariff. Quotas can avoid such recession induced imports.